PHYSICS MODULO GAUGE

The Methodology of the Incompleteness Framework

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ABSTRACT

The Incompleteness framework derives quantum mechanics, the Standard Model gauge structure, and a route to general relativity from a single starting point — that observation occurs — together with the structural conditions that an embedded observer must satisfy. This paper articulates the framework's *methodology*, separately from its physics, for readers in the foundations and philosophy-of-physics community. Five interrelated commitments organize the account. First, the framework's foundation, examined directly, resolves into a *two-axiom observation base*: an indubitable axiom (tokened differentiation occurs) and a substantive second axiom (differentiation recurs), the second presupposing but not derivable from the first — a no-go result we state and defend. Second, the framework realizes *constructive structural realism*: the equivalence class of structures consistent with the empirical inputs is uniquely determined, and the residual gauge freedom is explicitly classified, converting ontic structural realism from a programmatic position into a constructively demonstrated one. Third, the framework's relation to the no-go theorems of quantum foundations — Bell, Kochen–Specker, the Pusey–Barrett–Rudolph reality theorems, Frauchiger–Renner — is not piecemeal evasion but a single structural feature: each theorem assumes at the hidden-variable level a property that, in the framework, holds only at the observable level, and the trace-out construction is what mediates the two. Fourth, these combine into a general methodological proposal — that the well-posed question for fundamental physics is not "what is fundamental?" but "what is the equivalence class of structures producing observations, and what are its representatives?" — *physics modulo gauge*. Fifth, Part V develops the operational form of the methodology: a gauge-invariance no-go on substrate-only computation and a category-error finding on substrate-level Lagrangians sharpen the boundary between substrate-derivable content and content selected by empirical input; the two results unify with Part III's four cases under a single architectural pattern; systematic audits applying the operational standard to the framework's own content are summarized; and three methodological refinements canonized by the audit experience are stated at the generality the methodology requires. The paper develops each commitment and argues that, taken together, they constitute a replicable methodology rather than features peculiar to one framework. Developmental status is made explicit throughout: the foundational distillation of Part I is recent and is offered for scrutiny; the structural results of Part II are summarized from the framework's technical papers at the fidelity of faithful synthesis; Part III's four no-go cases

are all developed results — two (Bell, Frauchiger–Renner) summarized from the framework's technical papers, two (Kochen–Specker, Pusey–Barrett–Rudolph) developed in this paper as chains of reasoning on results those papers already establish; and Part V is more recent still — its two no-go results are developed in the framework's technical record, the systematic audits it summarizes are developed there as well, and the methodological reading as one architectural unification with Part III, together with the three operational refinements, is the present paper's synthesis, offered as a proposal in the same sense Part III's unifying formulation is offered.

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1. INTRODUCTION

1.1 What this paper is, and what it is not

This is a paper about method. The Incompleteness framework is a body of physics — it makes quantitative, parameter-free retrodictions across the Standard Model (including a value matching the JUNO neutrino measurement at 0.07σ) and derives the Bekenstein–Hawking entropy coefficient — and that physics is developed in a sequence of technical papers — [Main], [SM], [Substratum], [GR] — to which this one is a companion. The present paper does not re-derive any of it. It asks a different question: *what is the framework's methodology*, and is that methodology defensible and replicable independently of whether the framework's particular physical claims survive?

The question is worth separating because the framework's methodological posture is unusual in ways that a reader assessing the physics might not pause over, but that a reader in foundations or philosophy of physics will recognize as substantive. The framework claims a single foundational commitment and derives the rest. It claims that what other programs treat as free assumptions are, in its setting, forced. It claims a determinate relationship to the major no-go theorems of quantum foundations. And it claims that a long-standing question in the metaphysics of physics — the realism debate — is, as usually posed, the wrong question. Each of these is a methodological claim, evaluable on its own terms, and each is the subject of one part of this paper.

What this paper is not: it is not a defense of the framework's physics, which must stand on the technical papers and their empirical record; it is not a survey of the foundations literature, though it engages that literature where engagement is due; and it is not a finished statement. Part I in particular presents a recent distillation of the framework's foundational commitment, offered for scrutiny rather than asserted as settled. We mark developmental status explicitly wherever it applies.

1.2 The five commitments

The paper is organized around five methodological commitments, developed in Parts I through V.

Part I — the foundational commitment. The framework is often described as resting on a single axiom: *observation occurs.* Part I examines that phrase directly and finds it resolves into two axioms, not one — an indubitable claim that tokened differentiation occurs, and a substantive further claim that the differentiation recurs. Six of the framework's apparent further assumptions are recovered as lemmas of the first axiom; the second axiom is shown, by a no-go result, not to be derivable from the first. The methodological point is the discipline this exhibits: a foundation stated at the exact granularity of what it commits to, with the single non-evidential posit named and isolated. One element of the distillation is a commitment made explicit where it arises (§5): the observer's embeddedness — the proper-part decomposition $V \subsetneq S$ — is taken to belong to the primitive itself, because the framework reads "tokened differentiation" perspectivally (a registering from a locus, against a remainder). The thin alternative, on which embeddedness must instead be added as a third axiom, is noted there; the choice is presentational rather than empirical, and the two-axiom presentation reflects the adopted perspectival reading.

Part II — constructive structural realism. Ontic structural realism holds that physical structure, not individual objects, is what is fundamental and what is known. As usually stated it is a programmatic position. The framework makes it constructive: the equivalence class of structures consistent with the empirical inputs is shown to be uniquely determined, and the residual gauge freedom — the respects in which distinct structures are empirically indistinguishable — is explicitly classified. Structural realism becomes, in this setting, a demonstrated result rather than a stance.

Part III — per-theorem evasion of no-go results. The no-go theorems of quantum foundations — Bell, Kochen–Specker, the reality theorems, Frauchiger–Renner — each constrain hidden-variable accounts of quantum mechanics. The framework is a deterministic-substratum account and must answer to them. Part III shows the answer is not four separate escapes but one structural feature: each theorem assumes, at the level of the hidden substratum, a property that in the framework holds only at the level of the observable emergent description, and the trace-out construction that relates the two levels is what produces the failed assumption in each case.

Part IV — physics modulo gauge. The three preceding commitments combine into a general proposal. The realism debate asks what is fundamental; the framework's posture is that this question is ill-posed, and that the well-posed replacement is: what is the equivalence class of structures that produce the observations, and what are its representatives? Part IV articulates this as a methodology for fundamental physics generally, not as a feature peculiar to the framework's particular substratum.

Part V — the operational methodology. Parts II–IV establish physics modulo gauge as the right form of the answer; Part V develops what the methodology determines operationally. Two further no-go results sharpen the boundary between substrate-derivable content and content selected by empirical input: a gauge-invariance no-go on substrate-only computation, and a category-error finding on substrate-level Lagrangians. These two results unify with Part III's four cases under a single architectural pattern. Systematic audits applying the operational standard to the framework's own content are summarized, and three methodological refinements canonized by the audit experience are stated at the generality the methodology requires.

1.3 Developmental status, stated once

The five parts differ in maturity, and it is more honest to say so here than to qualify every paragraph. Part I is a recent distillation; its central no-go claim has been stress-tested but not yet refereed, and it is presented for exactly that scrutiny. Part II summarizes structural results — the uniqueness and gauge-classification theorems — established in the framework's technical papers, at the fidelity of faithful synthesis; a reader who needs the theorem statements themselves should consult those papers. Part III's no-go analysis treats four cases, all of them developed results: two — Bell and Frauchiger–Renner — established in the framework's technical papers, and two — Kochen–Specker and Pusey–Barrett–Rudolph — developed in this paper (Section 14), each as a chain of reasoning running on results the technical papers already establish. Part IV is a methodological proposal and is argued as such, not asserted. Part V is the most recent material; its two no-go results are developed in the framework's technical record, the systematic audits it describes are developed there as well, and the methodological reading of these as one architectural unification with Part III — together with the three operational refinements at the generality the methodology requires — is the present paper's synthesis, offered as a proposal in the same sense Part III's unifying formulation is offered. None of this paper's content is a derivation of physics; all of it concerns how the framework's physics is organized, and the claims of this paper stand or fall as methodology.

PART I — THE FOUNDATIONAL COMMITMENT

Part I develops the framework's foundational commitment: the distillation of "observation occurs" into a two-axiom base, the recovery of six apparent assumptions as lemmas, and the no-go result that isolates the single non-evidential posit. It is the first of the paper's four methodological commitments — the discipline of stating a foundation at the exact granularity of what it commits to.

2. THE EVIDENCE

2.1 What cannot be doubted

The framework invokes the Cartesian move: there is a fact that cannot be doubted, because the act of doubting it instantiates it. It is worth stating that fact in its most compressed form, because the compression is not the form the framework's definitions use, and the gap between them is the subject of this note.

The indubitable fact is **not** "I exist." "I" is a substantial noun; it imports a self, a persisting subject, a bearer of properties. It is not "an observer exists," for the same reason — "observer" is a relational substance-noun, importing both a thing that observes and a persistence through which it does so. It is not "the world exists" — that imports a totality.

What cannot be doubted, stated without importing any of these, is that **there is differentiated content**. There is some *this rather than that*. The registered content, whatever else is true or false of it, is not blank, not homogeneous: it has distinction in it. We will call this *tokened differentiation* — "differentiation" because the content is differentiated, distinction-bearing; "tokened" because what is in question is a concrete occurrence, an instance, not the abstract truth of a proposition. The abstract fact that seven differs from eight does not *occur*; what occurs is the concrete registering of a this-not-that.

This is the evidence. It is genuinely indubitable: to doubt it is to entertain the doubted against the affirmed, which is to register a distinction, which is an instance of the very thing doubted. It is also genuinely *thin* — thinner than "I exist," thinner than "a world exists." It asserts neither a subject nor a totality. It asserts that differentiation has occurred.

2.2 Why not thinner, why not thicker

One might propose a thinner floor still: "something exists." This is in fact *too* thin to be the evidence. Bare undifferentiated existence is not what is ever presented. What is presented is always some *this as against that* — structured content, not featureless being. Remove the differentiation and nothing remains to be aware of or to doubt. "Something exists" understates the evidence: the evidence includes the differentiation, not merely the existence.

One might propose a thicker floor: "the universe exists," "matter exists," "I exist." Each of these is more than the evidence. Each becomes the object of a coherent doubt the moment one asks how it is known; each, doubted, unravels to the thin fact beneath — that there is differentiated content. The thicker claims are not evidence. They are hypotheses about what lies behind the evidence.

Tokened differentiation is therefore the unique level that is *both* indubitable *and* the actual content of what is given: more than bare existence, less than a world, silent on every question of what underlies it. That it is also exactly thick enough to generate structure — as Section 4 shows — is not built in here; it is a fact established downstream, and it is what makes this level the right floor rather than an arbitrary one.

2.3 What the evidence is silent on

It is as important to state what tokened differentiation does *not* deliver as what it does.

It does not deliver a subject: there is differentiation, but the evidence does not say it is differentiation *for someone*, registered *by* an entity that has it.

It does not deliver a totality: there is differentiation, but the evidence does not say it is part of a larger whole, or that there is anything beyond it.

It does not deliver a ground: there is differentiation, but the evidence is silent on whether it is self-grounded, grounded in something else, or grounded in nothing. It asserts the occurrence and stops.

And — the point that will occupy Sections 3 and 5 — it does not deliver *time*. The fact "there is differentiated content" is, as stated, tenseless. It is satisfied by a single differentiated tableau: a this-here against a that-there, with no before and no after. The evidence gives a frame; it does not, on its own, give a sequence of frames.

2.4 The distinction-primitive: lineage and the divergence from it

Taking distinction as the primitive is not new, and the note should be explicit about its lineage. The major precedent is George Spencer-Brown's *Laws of Form* [SpencerBrown], which takes distinction as the irreducible starting point — "we cannot make an indication without drawing a distinction," and so "the form of distinction" is taken "for the form" itself. Spencer-Brown treats distinction as a proto-operation prior to all calculation and all representation: one cannot even compute $1 + 1$ without first distinguishing the symbols. The note's Axiom 1 is, in its core, the same recognition — that differentiation is the floor beneath which analysis does not reach.

But there is a deliberate and load-bearing divergence, and stating it precisely is part of what Axiom 1 contributes. Spencer-Brown's distinction is an *act*: the calculus begins with the *drawing* of a distinction, the *making* of a mark; in his treatment, as commentators note, there is no separation between the result of distinction and the *process* of distinguishing. That actival, processual reading is exactly what this note's Section 5.2 isolates as declined-attempt (1) — "distinguishing as act" — and rejects, because an act takes time and so smuggles succession into what is meant to be the tenseless floor. Axiom 1 therefore adopts Spencer-Brown's distinction-as-primitive but in a *tenseless, non-actival* form: it asserts that a distinction is *tokened* — concretely instantiated — not that a distinction is *drawn*. The difference is not cosmetic. It is precisely the difference that keeps Axiom 1 free of the temporal commitment that belongs, instead, to

Axiom 2. Where Spencer-Brown's single actival primitive already contains the seed of process, the note's two-axiom structure separates the tenseless content (Axiom 1) from the recurrence (Axiom 2) that an act would have fused into it.

This divergence also marks the limit of the borrowing. *Laws of Form* has been read, in the literature it generated, as a basis for claims about cybernetics, consciousness, and non-dualist metaphysics. The note takes from Spencer-Brown only the formal point — distinction as the irreducible primitive — and, consistent with Section 2.3, remains silent on the metaphysical and phenomenological superstructure that others have built on it. Axiom 1 asserts that a tokened differentiation occurs; it does not assert what the literature around *Laws of Form* has often gone on to assert.

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3. THE EIGHT-RUNG LADDER

3.1 The framework's definitions, unpacked

The framework formalizes "observation occurs" through a definition: an observation is a triple (S, φ, V) — a total system S, a dynamics $\varphi : S \to S$, and an observer $V \subsetneq S$, a proper subsystem with finitely many distinguishable internal states, coupled to its complement through φ ([Main] §1.2). Three lemmas and four conditions (C1–C4) are then attached. Laid against the evidence of Section 2, this formalization is seen to assert considerably more than "there is differentiated content." The surplus can be itemized as a ladder, each rung a distinct claim:

1. **A distinction occurs.** Differentiated content; this-not-that. *(The evidence of Section 2.)*
2. **The distinction is relational.** It holds between relata — a this *and* a that.
3. **The relata compose a whole the distinction does not exhaust.** There is an S, and the observer-locus V is a *proper* part of it: $V \subsetneq S$.
4. **There is more than one moment.** Temporal succession exists.
5. **A rule connects the moments.** There is a dynamics φ.
6. **The rule is bijective.** φ is deterministic (injective) and reversible (surjective).
7. **The states are finite in number.** The visible configuration space is finite.
8. **The hidden-sector conditions hold.** Non-zero coupling, slow-bath timescale separation, sufficient capacity, history readback (C1–C4).

The framework's theorem — that an embedded observer's description is necessarily non-Markovian, and quantum-mechanical for processes in the correspondence's class under (T) — takes all eight rungs as its basis. The evidence supplies only rung 1. The question of Sections 4 and 5 is which of rungs 2–8 are *consequences* of rung 1 and which are *additional commitments.*

3.2 The standard of derivation

A rung counts as derived — as a lemma rather than an addition — only if it follows from rung 1 by an argument that does not, at any step, presuppose a higher rung. The constraint is strict because the rungs are not independent: it would be circular to derive rung 4 using rung 5, or rung 6 using rung 7. Each derivation below is checked against this standard, and where a derivation consumes a premise not yet established, that is recorded as a finding rather than glossed.

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4. WHICH RUNGS ARE LEMMAS

4.1 Rung 2 — relationality

A distinction, by what the word denotes, holds between relata. A distinction that distinguishes nothing from nothing is not a distinction; the notion is incoherent without at least two terms. Rung 2 is therefore not an addition to rung 1 but an unpacking of it: "differentiation" already means *differentiation of something from something.* Rung 2 collapses into rung 1 by conceptual analysis.

4.2 The locus, and rung 3 — the proper part

Rung 3 asserts that the observer-locus is a proper part of an unexhausted whole. Between rung 2 and rung 3 there is a step easy to miss: rung 2 gives two relata symmetrically, while rung 3 gives an asymmetric part-whole structure, one relatum *contained in* a whole it does not exhaust. The step requires a premise — call it the *locus* premise — that the differentiation is registered *at* a locus, a where of the registering distinct from the what registered.

The locus premise is not a free addition. It is recovered from the verb in rung 1. To say a distinction *occurs* — as against merely *obtaining*, the way an abstract mathematical inequality obtains — is to say it is instantiated as a concrete token. And a concrete token is located: occurrence is indexical, a token is a *this-here*, instantiation places. The locus premise is therefore analytic in "occurs," just as relationality is analytic in "distinction." Once the locus is granted, rung 3 follows: the locus V, together with the registered content beyond it, composes a whole S of which V is a proper part, because V does not include the content beyond it. Rung 3 is a lemma.

This recovery has a substantive consequence. The proper-part structure $V \subsetneq S$ — the *incompleteness* the framework is named for — is not an independent postulate. It is a consequence of the evidence, provided "occurs" is read as "is tokened." Embeddedness is analytic in occurrence.

This recovery rests on a reading of the primitive, adopted here explicitly. The step from the locus premise to rung 3 depends on reading "tokened" as carrying a *perspectival* locus — a registering *from* a within-place, against a remainder.

On a thinner reading, "tokened" means only that the differentiation is concretely instantiated, with no within-place; on that reading the locus premise does not yield a *proper* part, and $V \subsetneq S$ would be a third axiom rather than analytic in the primitive. The framework adopts the perspectival reading — the floor the cogito secures is already a registering — so embeddedness is part of the primitive and the foundation is two axioms, not three. The choice is presentational: it changes no result downstream of the structure (S, φ, V). It is stated here rather than deferred, and rung 3 is presented as analytic in the perspectival primitive, with the thin-reading alternative noted.

4.3 Rung 5 — the existence of a dynamics

Granting (provisionally, pending Section 5) that there is more than one moment, and granting the locus that persists across moments as the same locus, there is necessarily *something* that relates the configuration at one moment to the configuration at the next. That relation is the dynamics φ. Rung 5 — the *existence* of a dynamics — is therefore a lemma, given rung 4 and the locus.

Two cautions. First, the lemma delivers only the existence of a successor relation, not its uniformity: that the same φ governs every locus and every moment is a further claim, recovered downstream from isotropy and homogeneity, not here. Second, the lemma depends on rung 4, which Section 5 will show is *not* itself a lemma. Rung 5 is thus a *conditional* lemma: a consequence of the evidence together with the one genuine addition.

4.4 Rung 6 — bijectivity

The framework's Lemma 3 derives bijectivity from the preservation of the observer's records. The derivation separates into two halves with different costs.

Determinism (injectivity). A registered distinction, once it has occurred, has occurred. If the dynamics φ merged two distinct total states into one successor — if φ were non-injective — then two configurations carrying distinct registered records would become a single configuration, and a distinction that had occurred would cease to have occurred. Rung 1 asserts the distinction occurs; persistence of the record across moments (rung 4 plus the locus) carries that occurrence forward; non-injective φ would violate it. So φ is injective. Determinism is a lemma, given rung 1, rung 4, and the locus.

The hidden-sector margin, closed. The record argument as just stated covers merges of configurations carrying distinct registered records. A merge of configurations differing *only in the hidden sector* erases no record, and needs a separate argument; there is one, and it is a dichotomy. If the merged configurations' successors differ visibly, the merge leaves a statistical trace: the marginal of a bijection under the uniform prior is doubly stochastic (Birkhoff–von Neumann; the framework's Main paper, §3.2), unitary emergent statistics are unistochastic, and a visibly-asymmetric merge breaks double stochasticity of the marginalized process at some step order — so such merges are excluded by the *observed unitarity* of quantum dynamics, an evidential input rather than a further axiom. If instead the merge leaves no statistical trace, it is removable: any map on a finite set restricts to a bijection on its eventual image, a recurrent registered history lives in that image, and the non-injective description is gauge-equivalent to a bijective one — the merge is a pre-history transient with no observational content, and the bijective representative is canonical (this is Part IV's surplus-as-inaccessibility, applied to the dynamics itself). Either horn delivers the bijective substratum for the dynamics governing registered history. The honest accounting, which the derivation's purity label should carry: injectivity on record-distinct configurations is a lemma of rung 1; injectivity on the hidden-sector margin is delivered by observed unitarity or by recurrence — a disciplined mixture of the evidential floor with one empirical input, not analysis of the evidential floor alone.

Reversibility (surjectivity). Injectivity yields bijectivity only on a finite set; on an infinite set an injective map need not be surjective. Reversibility therefore is not established here — it waits on rung 7.

4.5 Rung 7 — finiteness

Rung 7 asserts the visible configuration space is finite. The framework's Lemma 1 motivates finiteness from holographic entropy bounds — but those are downstream physical results, and invoking them at the foundational layer would be circular. The question is whether finiteness follows from rung 1 alone.

It does, in the form the framework actually requires. A *token* is a determinate concrete particular. A single tokened differentiation registers a determinate, and therefore finitely structured, content: an *infinite completed totality* of tokened distinctions is not something that can *occur*, because occurrence is the instantiation of a determinate particular and a completed infinity is not a determinate particular. Infinitely many distinctions could be *constructed* — one tokening after another, across moments — but never *given* in a single occurrence. The evidence therefore delivers finiteness *per moment*: each occurrence registers finitely many distinctions.

Finiteness-per-moment is exactly what the observation base delivers, and the scope of what it delivers should be stated on both of its axes. On the time axis: the visible configuration space $\mathcal{C}_V$ whose finiteness Lemma 1 requires is the space of states distinguishable *at the locus*, a per-moment object; the framework does not require, and should not claim, finiteness of the entire history, and an unbounded number of moments is consistent with the evidence. On the space axis: the tokening argument reaches only the *visible* factor — what the observer registers — so what is recovered as a lemma is per-moment finiteness of $\mathcal{C}_V$. The substratum arguments that act on the *total* per-moment space $\mathcal{C}_V \times \mathcal{C}_H$ — the Poincaré-recurrence step and the eventual-image restriction of the determinism recovery — consume finiteness of the hidden factor as well, and that extension is not delivered here: it is the structural assumption A1 of the substratum construction, counted as such in the companion paper's posit ledger ([Main §4.5], of which this rung analysis is the complementary view — the ladder counts what is recovered from the two-axiom base, the ledger counts what the full argument assumes). Rung 7, in the form Lemma 1 needs, is a lemma.

This completes rung 6 with the same scoping. Determinism (4.4) plus finiteness-per-moment yields bijectivity as a lemma on the visible factor: an injective map on a finite per-moment configuration space is surjective on it. Bijectivity of the substratum map on the total space inherits the A1 extension above; reversibility of the registered dynamics is recovered, and rung 6 is a lemma at the visible level, structural-plus-A1 at the substratum level.

4.6 Rung 8 — the hidden-sector conditions

Rungs C1–C4 are conditions on the relationship between the locus V and its complement. C1, non-zero coupling, follows conditionally: a distinction with content *about* the complement requires coupling to the complement, on pain of that content being unregisterable. C2 (slow-bath timescale separation) and C3 (sufficient capacity) are not lemmas — they are substantive conditions satisfied by some partitions and not others. The framework does not claim them as lemmas: it presents C1–C4 as the conditions defining *which* embeddings yield quantum mechanics ([Main] §3.4) and as a structural selection rule on where observers can exist ([Main] §4.6). Rung 8 is correctly treated by the framework as a selection condition, not a derivation, and no overclaim is present.

4.7 Summary of the recoveries

Of rungs 2–8: rung 2 (relationality) and the locus premise collapse into rung 1 by conceptual analysis; rung 3 (proper part), rung 5 (dynamics, conditionally), rung 6 (bijectivity), and rung 7 (finiteness-per-moment) are lemmas; rung 8 (C1–C4) is a selection condition the framework already treats as such. Exactly one rung remains: rung 4.

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5. THE ONE RUNG THAT IS NOT A LEMMA

5.1 Two questions about succession, and which one is the claim

The claim of this section concerns rung 4 — temporal succession. But "deriving succession" can mean two distinct things, and the no-go applies to one and not the other. The distinction is essential and the rest of this note depends on it.

The first question is the **ordering** question: *given* that there are multiple moments — multiple states of some persisting thing — can the relation "before/after" among them be defined without circularly invoking "before"? This is the problem the philosophy of time has worked on for a century. It is contested but not obviously hopeless: a reductive account of the ordering relation may well be possible (Section 5.3 discusses a recent candidate). The no-go of this note is *not* a claim about the ordering question.

The second question is the **domain-existence** question: is there a temporal domain *at all* — is there more than one moment, more than one state of a persisting thing — and can *that* be derived from a premise which does not already contain it? This is the question the no-go addresses, and the answer is negative.

The distinction matters because the two questions are routinely conflated, and an account that answers the first is easily mistaken for one that answers the second. An account of the ordering relation always *begins* from a set of moments, or a set of states of an object, or a configuration space — a multiplicity given at the outset — and derives the order *on* that multiplicity. It does not derive the multiplicity. The multiplicity is its input. And the multiplicity — that differentiation is instantiated across moments and not only within a single configuration — is exactly what the framework's second axiom asserts (Section 5.4).

Rung 4, as the framework needs it, is the domain-existence claim: that there is a temporal domain for the dynamics φ to act across. The claim of this section is that this cannot be derived from rung 1. Tokened differentiation, as established in Section 2, is tenseless: it is satisfied by a single differentiated tableau — a configuration, one frame — with no before and no after. To reach even *two* moments from a single differentiated frame requires a premise the frame does not contain. The argument takes the form of a small no-go result.

The two-question distinction also fixes the complete inventory of the framework's temporal commitments. The framework's downstream physics involves more temporal structure than the bare domain-existence claim — it involves an *ordering* of moments, a *causal topology*, and a derived *arrow* of time — and a reader is owed an account of how these relate to the two axioms. The inventory has four levels. *Multiplicity* — that there is more than one moment — is the domain-existence claim; it is the framework's second axiom. *Ordering* — that the moments form a sequence — is the ordering question above; the framework leaves it open. *Causal topology* — that the ordering carries influence-structure, that earlier moments can affect later ones — is what the dynamics requires to be stated at all. *The arrow* — that the ordering has an experienced direction — is derived in the framework's substratum treatment from horizon growth. The relations among these four are definite, and reduce the inventory to the two axioms-or-questions already named. The causal topology is not an independent commitment: an influence-structure on an ordered set of moments connected by φ is nothing over and above the ordering together with φ itself, since "earlier influences later" just means "related by forward iteration of φ"; the dynamics φ is independently present, so the causal topology reduces to the ordering. The arrow is derived, not posited, and rests on the causal topology, hence ultimately on the ordering. So the framework's genuinely independent temporal commitments are exactly two — multiplicity and ordering — and they are precisely the two questions this section distinguishes: multiplicity is settled by the second axiom, ordering is left open. Causal topology and the arrow are downstream of these two and of structure (the dynamics, horizon growth) the framework has independently. No third temporal axiom is required, and none is concealed: the two-question framing of this section is the complete account of where the framework's temporal commitments lie.

5.2 Six attempted derivations

The claim is established not by a single argument but by the exhaustion of the available derivations. Six attempts to derive succession from differentiation were constructed; each fails, and — the point of substance — each fails at a *distinct, identifiable step*, always the same kind of step: the illicit move from a state-predicate to a process-predicate, or from logical order to temporal order.

1. *Distinguishing as act.* "To distinguish is to do something, and doing takes time." Fails: it reads "distinguish" as an act, a process, when the evidence gives only differentiated content statically present. The frame need not be *arrived at.*

2. *Registering as record.* "To register is to inscribe a record, which persists to be read; reading is later than inscribing." Fails: it builds persistence — succession itself — into the meaning of "register." If "register" means only "differentiated content is present," no second moment follows.

3. *Actualizing as transition.* "To be actual rather than possible is to have been actualized, a transition from possibility to actuality." Fails: "transition" and "actualized" are temporal. Actuality can simply *be the case* tenselessly; the space of unrealized alternatives is a contrastive description, not a prior state the token passed through.

4. *Occurrence as actualization.* A refinement of (3) taking "occurrence" as the primitive. Fails at the same step: it still requires actuality to be the *outcome* of an actualizing, and "outcome" is temporal.

5. *Occurrence taking itself as object.* "Occurrence occurs — the occurrence of occurrence — and this self-application unrolls into a sequence." Fails differently: self-application generates a *logical* hierarchy of orders, not a *temporal* sequence. Reading the logical "of" as a temporal "then" is the added premise. (This attempt has a second defect: it imports a self-grounding, *causa sui*, thesis not present in the evidence, trading evidential austerity for elegance.)

6. *A single fused term.* The proposal that some sufficiently broad word denotes axiom and succession together as one. Fails: a single term denoting two separable contents does not unify them; it conceals the seam, so that

subsequent users inherit the succession commitment without being told they have made it. Concealment is not derivation.

Six attempts, six distinct points of failure, one invariant: succession is reintroduced, never derived. The consistency of the failure across genuinely different framings is the evidence that the obstruction is structural — a fact about what can be inferred from what — and not an artifact of any one bad formulation. A static premise does not yield a process conclusion; "differentiated content is" does not yield "differentiated content comes to be." In each case what cannot be reached is the domain-existence claim of Section 5.1: each attempt, to get a second moment, must assume a second moment.

5.3 The result against the existing literature

The no-go of Section 5.2 is not made in isolation from a substantial existing literature on temporal succession. That literature both corroborates the result and sharpens the distinction of Section 5.1.

Corroboration — Whitrow. G. J. Whitrow, in *The Natural Philosophy of Time* [Whitrow], surveyed the causal theorists' attempts to define temporal succession — the ordering of events by "before" and "after" — and found that they failed because they "tacitly invoked the very temporal order to be defined." Whitrow's verdict on Reichenbach's mark method, among the most ingenious such attempts, was the same. He concluded that "before" and "after" must be accepted as elementary. "Tacitly invoked" is precisely the diagnosis of Section 5.2: the attempts smuggle. That a major figure in the philosophy of time reached the same finding by independent survey is evidence the obstruction is real and not an artifact of this note's particular six framings.

The ordering question may be tractable — and that does not touch the no-go. It would overstate the result to claim the *ordering* relation cannot be reductively defined. A recent account (Wassermann [Wassermann]) proposes deriving "before" from a timeless "existence asymmetry" on the set of an object's realized states — a genuine candidate for answering the ordering question non-circularly. But the construction begins from M_X, the set of an object's actually-realized states. A set of multiple states of one persisting object is already a temporal domain: one object, many states, is exactly domain multiplicity. The account derives the *ordering* of those states; it assumes the *multiplicity.* This is not a criticism of the account — it answers the ordering question, which is the question it set out to answer. It is an illustration of the Section 5.1 distinction: even the most sophisticated reductive account of "before" takes the multiplicity as its starting set and cannot, by its own construction, derive it.

The same pattern in physics. The timeless approaches to quantum gravity exhibit the identical structure. Barbour's timeless physics [Barbour] takes "possible configurations of the universe and a law" as fundamental and derives time — but the configuration space, the multiplicity of possible instants, is primitive; what is derived is the ordering and the arrow. The Wheeler-DeWitt equation is timeless and time is recovered semiclassically — but on a space of geometries given at the outset. Even recent attempts to derive time from quantum entanglement have been observed to "use time in the set-up." Across both literatures — philosophy of time and the physics of timeless approaches — the same result holds: accounts derive the ordering of moments, or the arrow among them, from structures that already contain the multiplicity of moments. None derives the multiplicity. The domain-existence question is, in the words this note's Section 5.2 reached independently, not one that any known account answers.

Difference and repetition — Deleuze. The pairing of this note's two axioms — differentiation, and its recurrence — is the explicit subject of Deleuze's *Difference and Repetition* [Deleuze], the major modern treatment of the difference/repetition pair. The engagement matters because Deleuze's account of repetition is, at first sight, in tension with this note's Axiom 2. Deleuze insists repetition is "difference without a concept" — *not* the recurrence of the same but the production of novelty; and he ties repetition to the synthesis of time, citing Hume's observation that a repeated series (AB, AB, AB…) itself produces something new, a habit, an expectation. One might read this as a claim that recurrence is not a primitive but a synthesis, and that time is its product — which would make Axiom 2 a derived structure rather than an axiom. It does not. Deleuze's syntheses operate on an *already-given manifold* of instants: the AB,AB series already has its multiple terms; the passive synthesis of habit contracts a succession that is already there. What Deleuze derives — habit, expectation, the directedness from past through present to future — is temporal *ordering and arrow*, the Section 5.1 ordering level. He does not derive, and does not claim to derive, the bare existence of more than one instant. Deleuzian repetition therefore stands with Wassermann and Barbour: it presupposes the multiplicity and works upon it. It corroborates the no-go rather than threatening it. Two points of contrast are worth stating plainly. First, this note uses "recurs" in Axiom 2 in a deliberately thinner sense than Deleuze's "repetition": Axiom 2 asserts only that differentiation is instantiated more than once over, taking no position on whether the recurrence reproduces the same or produces novelty — that richer question is downstream of the bare domain-existence Axiom 2 asserts. Second, where Deleuze makes difference the sole ontological primitive and repetition a synthesis, this note finds two posits, not one, precisely because — as Section 5.2 establishes — the recurrence cannot be synthesized from the difference without a premise that already contains it.

5.4 The status of succession — a second axiom

Succession is therefore neither evidential nor derivable. But it is also not an independent axiom of the same kind as rung 1. The honest characterization is more specific.

Differentiation can be instantiated more than once over. There is differentiation *within* a configuration — this-here against that-there, the content of a single frame. And there is differentiation *across moments* — this-moment against that-moment. The principle invoked is the same in both: tokened differentiation. What the second instance adds is not a new principle but a new fact — that the principle is instantiated *recurrently*, more than once over, and not only within a single frame.

That fact is a second axiom. It is tempting to call it a mere parameter — a setting on the one axiom — or to absorb it by saying the one axiom is simply "applied across multiple domains." Both moves fail, and fail informatively. A posit that can be neither derived nor eliminated, and must be asserted, is an axiom, whatever it is called; "parameter" understates it. And "applied across multiple domains" is grammatically singular but contentually dual: application presupposes the existence of what one applies across, so the phrase silently carries the very second posit it appears to dissolve. The honest count is two.

The two axioms are:

- **Axiom 1.** *Tokened differentiation occurs.* Concrete differentiated content is instantiated; there is some this-not-that. This is the evidence of Section 2. It is indubitable — its denial is itself a tokened differentiation — and it is conceptually primitive: even the vocabulary needed to state the second axiom presupposes it.
- **Axiom 2.** *Differentiation recurs.* The principle of Axiom 1 is instantiated more than once over — not only within a single configuration but across a further axis. The framework's temporal domain is the minimal such recurrence; the configurational and temporal instances are the two minimum witnesses, and the actual count of axes is settled downstream by derivation, not fixed here. What Axiom 2 asserts is bare recurrence — that the principle holds more than once over — and nothing more: it does not by itself supply an ordering of the recurrent instances, a causal topology, or the specific dynamics φ; these are either left open (ordering, §5.1) or derived downstream from structure the framework independently has (causal topology and arrow, §5.2 inventory). The floor is two posits at the granularity stated, and the dynamical content the downstream physics requires enters through later, separately-stated structural conditions rather than through Axiom 2's semantics.

The two axioms stand in a definite relationship, and stating it precisely is part of the result. Axiom 2 **presupposes** Axiom 1: there can be no recurrence of differentiation without differentiation, and the very notion of an "axis" or "domain" along which the recurrence runs is itself a differentiation-structure, unstatable without Axiom 1's content. But Axiom 2 is **not derivable from** Axiom 1: that a differentiation occurs does not yield that it recurs — this is the no-go of Section 5.5. The relationship is therefore one-directional presupposition without entailment. Axiom 1 is first — conceptually and epistemically. Axiom 2 is genuinely additional. They are two, ordered, and not reducible to one.

A note on what is *not* claimed. The two axioms are not unrelated posits. The principle they invoke — tokened differentiation — is single; Axiom 2 is that single principle asserted to recur. The unity is real, but it is a unity of *theme*, not of *count*: one principle, two axioms. To let the thematic unity be reported as a unity of count — "one axiom" — would be to conceal the second posit behind the first, which is the move this distillation exists to expose, not to perform.

5.5 The result stated as a no-go

The content of Sections 4 and 5 can be stated as a single result. *Given Axiom 1, "tokened differentiation occurs," the rungs relationality, proper-part, dynamics-existence, determinism, finiteness-per-moment, and bijectivity follow as lemmas; the recurrence of differentiation — Axiom 2, equivalently the existence of more than one moment for the dynamics to act across — does not, and cannot, follow.* The negative half is a no-go theorem of modest scope but real content: it identifies, exactly, the one place where the framework's foundation makes a second commitment, and it shows that the second commitment cannot be discharged by any derivation from the first — it must be asserted as an independent axiom. The no-go is specifically about the *recurrence* of differentiation (the existence of the temporal domain), not about the *ordering* of moments within that domain; the latter is a separate and possibly tractable question (Section 5.3), and the framework takes no position on it here.

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6. THE TWO-AXIOM OBSERVATION BASE

6.1 Statement

The foundation of the framework, distilled, is two axioms:

> **Axiom 1.** *Tokened differentiation occurs.* — Concrete differentiated content is instantiated: there is some this-not-that. Indubitable; conceptually primitive.
>
> **Axiom 2.** *Differentiation recurs.* — The principle of Axiom 1 is instantiated more than once over, across a further axis. The temporal domain is the minimal such recurrence; configurational and temporal instances are the two minimum witnesses, the actual count of axes being settled downstream by derivation.

We call this two-part foundation the **two-axiom observation base**. The two axioms are ordered: Axiom 2 presupposes Axiom 1 (recurrence of differentiation requires differentiation, and the very notion of an axis along which recurrence runs is a differentiation-structure), but Axiom 2 is not derivable from Axiom 1 (Section 5.5, the no-go). The relationship is one-directional presupposition without entailment.

The foundation comprises two axioms, not one axiom plus a domain-multiplicity *parameter*: a posit that can be neither derived nor eliminated, and must be asserted, is an axiom, so the count is two. The two axioms invoke a *single* *principle* (tokened differentiation), Axiom 2 being that principle asserted to recur. The unity is thematic, not numerical — one principle, two axioms — and is recorded as the relationship between the axioms, not as a reduction of their count.

Everything else in the foundational layer is a lemma: relationality, the locus, the proper-part structure $V \subsetneq S$, the existence of a dynamics, determinism, finiteness-per-moment, and bijectivity. The hidden-sector conditions C1–C4 are selection conditions, not lemmas, and are so treated.

6.2 What the two-axiom form buys

The two-axiom statement is more honest than "a single axiom" and more precise than "several assumptions," and both improvements are substantive.

Against "a single axiom": that phrasing, unqualified, suggests the framework's foundation is a single indubitable fact. It is not. It is one indubitable fact (Axiom 1) *plus* one further commitment (Axiom 2) that is not indubitable — a world without recurrence, a single tenseless configuration, is coherently conceivable, merely a world with no dynamics and no physics. The two-axiom form states this. It does not weaken the framework: a foundation whose second, non-indubitable commitment is named, isolated, and shown by a no-go theorem to be irreducible to the first is more defensible than a foundation whose commitments are distributed invisibly through a definition. The framework need not claim "one axiom"; the true claim — two axioms, ordered by presupposition, provably not collapsible — is the stronger thing to be able to say, because it comes with a theorem rather than a slogan.

Against "several assumptions": the eight-rung ladder might suggest the framework rests on eight independent posits. It does not. Six of the eight are lemmas. The framework rests on two axioms, and the two-axiom form states *that* — the economy is real, and naming it at the correct count is part of stating it honestly.

6.3 Relation to other minimal claims

The two-axiom observation base is sometimes compared, in discussion, to other candidate "bedrock" claims. The comparisons are clarifying. They concern Axiom 1 — the indubitable, conceptually primitive member — since it is Axiom 1 that occupies the "bedrock" role.

It is not "the universe exists." That asserts a totality; Axiom 1 asserts only tokened differentiation, and the totality (S, the unexhausted whole) is recovered as a lemma (rung 3), not assumed.

It is not "a being exists" — in particular not any claim of the form "X exists" for X a substance with properties. The distillation refuses substance-nouns throughout: it rejected "I exist" and "an observer exists" for importing a persisting subject, and rejected "occurrence grounds itself" for importing a self-grounding thesis. Axiom 1 asserts an occurrence and is silent on what, if anything, underlies it. It is ground-neutral by construction, and that neutrality is deliberate: it is the same discipline that rejected attempt (5) of Section 5.2.

It is not "something exists" simpliciter. That is too thin to be the evidence (Section 2.2): bare undifferentiated existence is not what is given. Axiom 1 asserts *differentiated* content — more than bare existence — which is both what is actually given and, as Section 4 shows, exactly enough structure to generate the lemmas.

Axiom 1 belongs to the family of claims that cannot be doubted because doubting instantiates them — the family of the Cartesian *cogito*. It differs from the others in that family by being maximally disciplined about claiming *only* the indubitable floor and not the structure that the tradition has usually built onto it without notice. Axiom 2, by contrast, makes no claim to indubitability; it is a substantive posit, defended not as undeniable but as the minimal further commitment without which there is no dynamics and no physics, and shown by the no-go to be irreducible to Axiom 1.

6.4 Relation to the landscape: generality, selection, and what they presuppose

A natural question, given that the framework elsewhere derives a gauge structure and a selection principle, is whether the two axioms might themselves be understood on the model of the string-theory landscape: the laws are not unique but range over a vast space of consistent possibilities — gauge or moduli freedom in the loose sense — and *which* possibility obtains is fixed, in our case, by anthropic selection, since only observer-permitting possibilities are observed. Could Axiom 2 — the recurrence of differentiation, the existence of the temporal domain — be such a selected feature: gauge in general, specific in our anthropic case?

It could not, and the reason is the same no-go established in Section 5, arriving from a new direction. Anthropic selection on a landscape operates on features *within an already-constituted physics*: the cosmological constant, the compactification, the couplings each take different values across the landscape, but every point of the landscape is a temporally-extended world with a configuration space and a dynamics. The landscape is a landscape *of* differentiated, temporally-extended things. Anthropic selection picks among such worlds; it cannot pick whether there is a temporal domain at all, because the selection apparatus — a measure over possibilities, the counting of observer-permitting cases — already presupposes the temporal domain. A "world" with no recurrence of differentiation has no dynamics, no vacua, nothing to be a point of a landscape and nothing for a selection measure to weigh. The proposal "Axiom 2 is an anthropically-selected landscape variable" therefore relocates the second commitment into the meta-level structure of the landscape rather than removing it: the landscape itself presupposes Axiom 2. This is the same outcome as the "parameter" and "condition" framings of Section 5.4 — a setting is a setting *within* a space of possibilities, and the space of possibilities cannot be constituted without the very structure the setting was meant to make optional. It may be recorded as a further attempt — alongside the six of Section 5.2 — examined and found to presuppose what it would explain.

String theory itself bears on the two axioms only in this presuppositional way. It is formulated on a worldsheet with a time coordinate, in a target space with a time dimension, over a configuration space that is a multiplicity; it inherits both Axiom 1 and Axiom 2 and questions neither. String theory is not a competing account of the foundation discussed here — it operates well downstream of it, and whatever its landscape is, it is a landscape of worlds in which both axioms already hold.

The genuine and useful point in the landscape comparison lies *below* the two axioms, not at them. Downstream of Axiom 1 and Axiom 2, the framework does have a "general, then selected" structure, and it is one of its established features rather than a speculation: the emergent description is gauge — partition-relative, with the substratum gauge group identifying which structure is observable — and the partition that supports an observer is then *selected*, by the structural observer-selection theorem, as one of those satisfying the embedding conditions. This is the framework's analogue of "general law, particular vacuum"; it is the natural counterpart of the landscape-plus-anthropics picture. The difference worth stating is that the framework's version is derived: where the landscape picture posits a space of vacua it cannot derive and an anthropic measure it cannot rigorously define, the framework's gauge structure is constructively classified and its selection principle is a theorem about which partitions admit embedded observers. The "gauge in general, specific in our case" intuition is therefore correct and is realized in the framework — but it is realized one level down, among the partitions and the emergent descriptions, not at the two axioms, which are precisely what must hold for there to be a space of partitions to be general over in the first place.

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7. THE INTUITION IN PLAIN LANGUAGE

The preceding sections are precise at the cost of being technical. The underlying intuition is simple, and stating it plainly is worthwhile — both because the simplicity is the point and because a reader who holds the intuition can check the technical claims against it.

Begin with the one thing that cannot be doubted. Not "the world is real," not "I exist," not that anything in particular lies behind appearances — those are already guesses about what is behind it. Strip them away. What remains, before any guessing, is this: there is something rather than nothing, and not a blank something but a something *with a difference in it.* This, not that. The fact that things are not all the same.

That is the whole starting point. Not “a universe exists” — that already assumes one great container holding everything. Not “a God exists” — that assumes a thing that has properties and does things. Just: *difference shows up.*

It is the right floor, and not a thinner or thicker one, for a specific reason. Thinner — “something exists” — is too thin: plain existence with no difference in it is not anything you are ever actually given, and nothing could be aware of it or doubt it. Thicker — “a universe,” “matter,” “I” — is smuggling: each sounds obvious only because one already lives inside the assumption, and each, once questioned, dissolves back to the thin fact underneath, that there is difference.

Now the second piece, the piece that took the most care to get right. There are two ways difference can show up, and they are genuinely different. Difference *across a scene*: this thing here is not that thing there — two things side by side, in one frozen snapshot. And difference *across time*: this moment is not that moment — one thing, but later.

One might hope the second follows free from the first — that “there is difference” hands you time automatically. It does not. A single frozen photograph is full of difference — light here, dark there, edges, shapes — and contains no time whatsoever; nothing in it is before or after anything else. So “there is difference” gives the photograph. It does not give the movie. To get the movie — time, change, one-thing-then-another — one must add something: that difference runs in the time direction too, not only across the scene. And that addition cannot be squeezed out of the first claim. Six ways of trying to squeeze it were attempted; every one smuggled time in through a word that already had time hidden inside it.

So the honest foundation is two things of two different kinds. The first is *known*: there is difference — it cannot be doubted, because doubting is itself a telling-apart. The second is *assumed*: the difference runs in the time direction as well. The framework does assume the second — plainly our world has time — but it is honest that this is a thing put in, not a thing found.

In one sentence: *“there is difference” is what we know; “the difference includes time” is the one thing we assume; physics is what follows from the two — and being able to point at exactly that one assumed piece is what makes the foundation honest.*

That last clause is the reason the exercise matters. Most attempts at a foundation for physics begin, without noticing, from “there is a world, with things in it, and time passing,” and never identify how many separate commitments that involves. The two-axiom observation base says: one thing is indubitable, one further thing is assumed, and here is precisely which is which. A foundation in which the single assumed brick can be pointed at is sturdier than one in which the assumptions are smeared invisibly through everything — not because it assumes less in total, but because it is honest about what it assumes.

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8. OPEN QUESTIONS AND STATUS

Part I is developmental, and several questions are deliberately left open.

The succession no-go. The central negative result — that the existence of the temporal domain is not derivable from tokened differentiation — rests on the exhaustion of six derivation attempts (Section 5.2), corroborated by the independent verdict of the philosophy-of-time literature (Section 5.3). Six is not infinity. The result would be overturned by a seventh attempt that succeeds: a derivation of the temporal domain — more than one moment — from a tenseless differentiation premise that does not, at any step, smuggle a process- or temporal-order predicate, and does not begin from a multiplicity (a set of states, a configuration space) that already contains the domain. The author has not found such a derivation and has given a structural reason to expect none exists — the state-to-process gap is not the kind of gap an argument crosses — but the question is left open for scrutiny. A philosopher of physics is the natural reader to either close it or confirm its standing. The distinction of Section 5.1 should be kept in view in any such scrutiny: a reductive account of the *ordering* of moments (Section 5.3 notes a recent candidate) does not bear on the no-go, which is about domain-existence.

The count, and the relationship between the axioms. Part I settles on two axioms — Axiom 1 (tokened differentiation occurs) and Axiom 2 (differentiation recurs) — related by one-directional presupposition: Axiom 2 presupposes Axiom 1 but is not derivable from it. Earlier formulations that sought a single-axiom statement (a "parameter," a "condition," a principle "applied across domains," a reverse derivation of Axiom 1 from a domain-multiplicity axiom, or an anthropically-selected landscape variable — Section 6.4) were each examined and withdrawn; each, on inspection, either understated Axiom 2's status or relocated the second commitment without removing it. The two-axiom reading is the current resting point. It is offered, like the rest of Part I, as developmental: whether the relationship between the axioms admits a still more economical regimentation is left open, but no single-axiom formulation found so far survives scrutiny, and the burden is on any proposed one to show it does not conceal the second posit.

What is not open. Two points are not left open. First, the lemma recoveries of Section 4 — relationality, the locus, the proper-part structure, dynamics-existence, determinism, finiteness-per-moment, bijectivity — are presented as established, given the axiom; they turn on conceptual analysis of "occurs" and "tokened" and are checkable directly. Second, the distillation has no bearing on whether the framework's downstream derivations succeed; it concerns only the formulation of their starting point.

PART II — CONSTRUCTIVE STRUCTURAL REALISM

Part II develops the framework's second methodological commitment: that ontic structural realism, ordinarily a programmatic stance, becomes in this framework a constructively demonstrated result. Two theorems carry the claim — a reconstruction theorem and a gauge-classification theorem — and the methodological point is what their conjunction makes possible: a realism about structure that is earned by construction rather than asserted as a philosophical position. The theorem statements summarized here are established in the framework's substratum paper; this part states them at the fidelity of faithful synthesis and refers the reader needing the proofs to that paper.

9. STRUCTURAL REALISM AS A PROGRAMMATIC POSITION

Ontic structural realism (OSR) holds that what is fundamental, and what physical theory gives us knowledge of, is structure — the network of relations — rather than individuals bearing intrinsic natures. The distinction between OSR and a merely epistemic structural realism was drawn by Ladyman [Ladyman1998], and the ontic position has been developed principally by Ladyman and Ross [LadymanRoss] and by French [French2014], with the relation to quantum physics worked out by French and Ladyman [FrenchLadyman2003] and a recent "effective" refinement given by Ladyman and Lorenzetti [LadymanLorenzetti]. The position emerged from two pressures: the pessimistic meta-induction, which observes that the theoretical objects of successive physical theories are repeatedly discarded while their structural-mathematical content is retained; and the underdetermination of object-level descriptions by structure, sharpest in physics where distinct object assignments (particles-as-individuals versus particles-as-modes, for instance) leave all relations and all predictions intact.

OSR responds: take the structure as the ontology, and the recurring instability of the object level becomes unsurprising, because the object level was never the bearer of the theory's content. A standing concern for the position, recognized in its own literature, is that "structure" not collapse into "pure mathematics" — French and Ladyman were explicit that OSR is not committed to the world being purely mathematical — and the question of how the structural and the physical are related remains under discussion. The framework's treatment, developed below, bears on exactly this concern: it makes the structure a specified mathematical object while keeping the physical content fixed by empirical inputs, and it is explicit (Section 12) about what is thereby claimed and what is not.

As ordinarily developed — in the work of Ladyman, French, and others — OSR is a *programmatic* position. It is a stance one adopts about physical theory in general, defended by general arguments (the meta-induction, the underdetermination), and it makes a general promise: that for a mature physical theory, the structure could in principle be specified and the object-level residue shown to be surplus. The promise is rarely discharged for any specific theory. What "the structure" is, precisely; which respects of a theory's description are structural and which are surplus; whether the structure is uniquely fixed by the empirical content or itself underdetermined — these are, in the programmatic version, left as the position's horizon rather than its deliverable.

The methodological claim of this part is that the Incompleteness framework discharges the promise. It does so not by arguing for OSR as a stance but by proving two theorems whose conjunction *is* the structural-realist claim made constructive: the structure consistent with the empirical inputs is exhibited and shown to be unique, and the surplus — the respects in which distinct descriptions are the same structure — is exhibited and shown to be exhaustively classified. OSR, in this framework, is not adopted. It is a result.

10. THE RECONSTRUCTION THEOREM: UNIQUENESS OF THE STRUCTURE

The framework's forward direction runs from a finite deterministic construction — a bijection (S, φ) on a finite configuration space — to observed physics: the emergence of quantum mechanics for an embedded observer, the Standard Model gauge group as the visible-sector projection, the gravitational sector from boundary thermodynamics. The reconstruction theorem addresses the converse. It asks whether observed physics determines the construction, and answers that it determines it uniquely, up to a gauge equivalence that the next section classifies.

Stated as faithful synthesis of the framework's substratum paper [Substratum]: given empirical inputs E1–E7 — quantum mechanics with Bell-inequality violations attaining the Tsirelson bound, finite boundary entropy, spatial isotropy, propagating gravity, stable matter, boundary-entropy concordance — and structural assumptions A1–A6 — finiteness, determinism, bounded coupling degree, center independence, linearity, background independence — the equivalence class $[(S, \varphi)]/\mathscr{G}_{\mathrm{sub}}$ is uniquely determined. The reconstruction yields, as forced outputs rather than fitted parameters, a finite set with a bijection of bounded coupling degree and statistical isotropy; spatial dimension three; the coupling-matrix eigenvalue multiplicities $(3, 2, 1)$; the Standard Model gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$; three generations (physical reading conditional on H-spin', [SM §4.7]; chirality on H-χ', [SM §4.8]); one Higgs doublet; anomaly-free hypercharges; the vanishing of the strong-CP angle, $\bar{\theta} = 0$; and the relation $\hbar = c^3\epsilon^2/(4G)$ tying Planck's constant to the discretization scale and the gravitational constant.

Two features of this theorem matter methodologically, independently of whether each forced output survives empirical scrutiny.

The first is the *direction* of the result. The theorem does not say that the framework's construction can reproduce observed physics — reproduction is the forward direction, and many constructions can be tuned to reproduce a target. It says the inverse: that observed physics, taken as the input, *backs out* to the construction with no freedom left except the classified gauge equivalence. This is the property an OSR claim needs and ordinarily cannot supply. To say "the structure is what is real" is idle if many structures fit the phenomena; it has content exactly when the structure is pinned. The reconstruction theorem is the pinning.

The second is the theorem's *honesty about its inputs.* The structural assumptions A1–A6 are not claimed as derived; they are restrictions on the class of constructions the reconstruction considers, and the framework states this. The uniqueness is uniqueness *within* the A1–A6 class. This is not a concession that weakens the result so much as a specification of its scope: the theorem says that, holding fixed a small set of explicitly stated structural assumptions, the empirical inputs determine the structure. A reader who rejects an assumption in A1–A6 is not shown a contradiction; they are shown which alternative they must develop. The methodological discipline — every non-derived commitment named and exposed — is the same discipline Part I exhibited at the foundational layer, here operating at the level of the reconstruction.

There is also a scope limit the framework states itself, and faithful synthesis requires repeating it. The empirical inputs E1–E7 are idealized: they are structural facts about observed physics — the full apparatus of quantum mechanics, Bell violations exactly at the Tsirelson bound, the holographic bound, exact isotropy — none of which any finite body of data establishes outright. The theorem proves uniqueness under the idealized inputs. Reconstruction from finite data is a distinct and weaker question, on which the framework's answer is the cumulative weight of its quantitative retrodictions rather than the theorem. Part II's claim is about the theorem's idealized uniqueness; it does not assert that finite data singles out the structure, and a referee should hold it to the former and not the latter.

11. THE GAUGE-CLASSIFICATION THEOREM: EXHIBITING THE SURPLUS

A reconstruction theorem alone would establish uniqueness up to *something* — up to whatever transformations leave the empirical inputs invariant. An OSR claim needs that "something" itself to be exhibited and bounded. Otherwise the surplus is a residual unknown, and "the structure is real, up to we-know-not-what" is not yet a constructive position. The framework's second theorem supplies this: it classifies the gauge group $\mathcal{G}_{\mathrm{sub}}$ — the transformations relating distinct constructions that produce identical observables — into an explicit and provably exhaustive list.

As faithful synthesis: the substratum gauge group $\mathcal{G}_{\mathrm{sub}}$ is generated by four families of transformations. *State relabeling* — conjugating the bijection by any permutation of the configuration space; transition probabilities depend on the coupling structure, not on the labels, and this family alone is larger than any Standard Model gauge group. *Alphabet change* — replacing the local state space $\mathbb{Z}/q\mathbb{Z}$ by $\mathbb{Z}/q'\mathbb{Z}$ for any $q, q' \geq 2$, with every prediction proved independent of the choice. *Deep-sector enlargement* — adjoining hidden degrees of freedom beyond the boundary layer, with arbitrary slow dynamics, leaves the emergent description unchanged, the observables depending only on the visible and boundary sectors. *Graph isomorphism up to statistical isotropy* — two coupling graphs with the same polynomial growth dimension, spectral mode count, and large-scale isotropy produce the same emergent quantum-mechanical, dimensional, and gravitational content, so for that content the regular cubic lattice and a suitable bounded-degree random graph are gauge-equivalent. The Standard Model gauge group is the qualification: it depends on the octahedral point group of the cubic lattice (the (3,2,1) multiplicities), which a generic isotropic graph does not supply, so there the generator is restricted to cubic-structure-preserving isomorphisms and the simple-cubic Bravais type is an empirically-selected input (see [Substratum §4, Theorem 24(iv)]).

The decisive word is *exhaustive*. The framework does not merely list four families it has noticed; it argues their completeness. For the state-channel generators (state relabeling and deep-sector enlargement) the argument is routed through the uniqueness clause of Stinespring's dilation theorem: two dilations of the same visible-sector channel differ only by hidden-sector transformations of exactly the kinds those families enumerate. The graph-isomorphism family (iv) is not closed by Stinespring uniqueness — it acts on the coupling-graph structure that feeds the gauge derivation downstream of the visible-sector channel — and is established only for the emergent-QM/dimensional content, with the cubic-structure qualification of [Substratum §4, Theorem 24(iv)] for the gauge sector; its completeness for the gauge content does not follow from Stinespring uniqueness. Candidate fifth families — time reversal, hidden-sector dynamical reparametrization — are shown to be already contained in the four. The surplus is not just exhibited; it is closed.

That a gauge freedom can be exhibited and bounded is worth contrasting with a known difficulty in the foundations of quantum field theory. In QFT with infinitely many degrees of freedom, the existence of unitarily inequivalent representations means that the choice of representation is a substantive interpretive problem rather than a settled one — a problem Ruetsche's analysis of interpreting quantum theories develops at length [Ruetsche]. The framework's setting is finite, and the gauge-classification theorem is what the finiteness buys: the analogue of "which representation" is "which representative within the equivalence class," and that question is not left as an open interpretive choice but answered by an exhaustive classification. This is not a criticism of the QFT situation, which concerns genuinely infinite systems; it is a statement of what the framework's finiteness makes available — a closed classification where the infinite case admits none.

This is what converts the structural-realist claim from programmatic to constructive. The programmatic version says: there is a structure, and object-level descriptions carry surplus. The framework's version says: *here* is the structure — the equivalence class fixed by the reconstruction theorem — and *here*, in four enumerated and provably complete families, is exactly the surplus. The two theorems together leave nothing in the position's promissory note. What is real is the equivalence class; what is gauge is the four-family freedom; and which specific construction within the class one's universe "is" is — provably — without empirical content, because the gauge group relates all representatives to identical observables.

12. WHAT THE CONJUNCTION ESTABLISHES, AND WHAT IT DOES NOT

The methodological yield of Parts II is the conjunction, so it is worth stating exactly.

With both theorems in hand, the framework can say three things that programmatic OSR cannot. It can *exhibit* the structure rather than promise it: the equivalence class is a specified mathematical object. It can *bound the surplus*: the four-family classification says precisely which differences between descriptions are differences that make no difference. And it can *adjudicate the realism question for its own case*: realism about a specific substratum is not false but unstatable — it commits to facts the framework's own gauge theorem classifies as observably empty — while a deflationary reading on which the framework is merely an economical reformulation undersells the reconstruction theorem, since uniqueness means the structure is not one economical option among many but the structure the evidence forces. The framework's position is the middle one: realism about the equivalence class, exactly.

Three honest limits. First, the theorems are summarized here, not proved; their proofs and their dependency structure live in the framework's substratum paper, and the substratum paper itself flags that the reconstruction theorem's confidence is bounded by the confidence of its dependencies. Part II inherits that flag and does not launder it. Second, the constructive-OSR claim is conditional on the structural assumptions A1–A6, which are not themselves derived; the claim is "constructive OSR within the A1–A6 class," and the qualifier is not optional. Third, the claim is about the framework's particular construction. That a constructive OSR is achievable *here* does not show it is achievable for physics in general — but it does show the programmatic position has at least one constructive instance, which is itself a contribution to the OSR literature: an existence proof that the promissory note can be paid.

It is worth noting that the *idea* of reconstructing structure from observables is not foreign to mathematical physics. Algebraic quantum field theory contains a reconstruction result of its own — the Doplicher–Roberts theorem, by which the field algebra and the global gauge group are recovered from the category of physical representations of the observable algebra [Halvorson]. The framework's reconstruction theorem is not that theorem and does not depend on it, but the precedent matters methodologically: it shows that "the observable content determines the structure, including a gauge group, up to a specified equivalence" is a known *form* of result in rigorous physics, not a novel species of claim. What the framework adds, relative to that precedent, is the scope — a reconstruction not of a field algebra's gauge group but of a complete substratum-equivalence-class from which the emergent quantum description, the Standard Model gauge group, and the gravitational sector all follow — and, with it, the exposure to a correspondingly larger set of structural assumptions, which is why the A1–A6 qualifier is load-bearing.

The methodological moral, which Part IV will generalize, is already visible. Once the structure is pinned and the surplus is classified, the question "but which substratum is the real one?" stops being a deep unanswered question and becomes a question shown to have no content. That dissolution — not the answering of the question but the demonstration that it was ill-formed — is the pattern the framework applies again, in Part III, to the no-go theorems of quantum foundations.

PART III — PER-THEOREM EVASION OF NO-GO RESULTS

Part III develops the framework's third methodological commitment: that its relation to the no-go theorems of quantum foundations is governed by a single structural feature rather than by four separate escapes. The Bell and Wigner's-friend analyses summarized here are established in the framework's Main paper; the Kochen–Specker and reality-theorem cases are developed in this part, each as a chain of reasoning running on results the technical papers already establish. The unifying account — that the framework's two-level, trace-out architecture is what each theorem's assumption runs aground on — is the present paper's synthesis, and the developmental status of each case is flagged accordingly.

13. THE NO-GO THEOREMS AS A FAMILY

A deterministic-substratum account of quantum mechanics must answer to the no-go theorems of quantum foundations. These results — Bell's theorem [Bell], the Kochen–Specker theorem [KochenSpecker], the reality theorems in the Pusey–Barrett–Rudolph family [PBR], the Frauchiger–Renner theorem [FrauchigerRenner] — each show that some intuitively natural class of accounts of quantum mechanics cannot reproduce its predictions. A framework that posits a deterministic substratum underneath quantum mechanics is, on its face, exactly the kind of account these theorems constrain, and a reader encountering such a framework is right to ask, immediately, how it survives them.

The framework's answer could in principle be piecemeal: a separate manoeuvre for each theorem, each manoeuvre exploiting whatever assumption that particular theorem happens to leave open. Piecemeal answers are methodologically weak — they invite the suspicion of ad hoc construction, a different escape hatch cut for each prison — even when each individual answer is technically correct. The methodological claim of this part is that the framework's answer is not piecemeal. The four theorems are met by one structural feature of the construction, and the appearance of four separate evasions is dissolved into a single observation about levels of description.

The single observation is this. Each of these no-go theorems assumes some property — locality and factorizability, or noncontextuality, or a form of state-realism, or the universality of a single quantum description — and shows that no account having that property at the level of its hidden or underlying variables can reproduce quantum mechanics. The framework has two levels of description — a hidden deterministic substratum and an emergent quantum description — joined by the trace-out, the marginalization of the substratum dynamics over the unobserved hidden sector that produces the emergent description in the first place. The framework's relation to the whole family of theorems is mediated by that one piece of machinery, already at its core, and fixed before any no-go theorem is considered.

The trace-out separates the two levels, and that separation is what each theorem's assumption runs aground on — though not in identical form for all four, and the differences are worth stating precisely rather than smoothing over. For three of the theorems the pattern is the same: the assumed property — Bell's factorizability, Frauchiger–Renner's single universal description, PBR's preparation-independence — holds at the emergent level, where it is unobjectionable because that level simply is quantum mechanics, and fails at the substratum level, where the theorem needed it. For the fourth, Kochen–Specker, the property — a noncontextual valuation of the observables — holds at *neither* level: the emergent level is quantum mechanics, which by Kochen–Specker's own content has no noncontextual valuation, and the substratum level has a definite configuration that is not a valuation of the emergent observables at all. What the four cases share is therefore not a single "emergent-yes, substratum-no" template but the architecture itself: the trace-out, by interposing a partition-dependent marginalization between substratum and emergent observable, is what makes the noncontextual valuation unavailable in the KS case and what places the property at the emergent level only in the other three. One architecture; four manifestations of working within it. The theorem's "no account can do this" is met, in each case, by locating exactly where the framework's two-level structure declines to supply the object the theorem's assumption is about.

14. THE FOUR CASES

A status note, stated before the cases and not buried within them. All four cases below are developed results, not conjectures. Two — Bell and Frauchiger–Renner — are developed in the framework's Main paper and are summarized here *faithfully. The other two — Kochen–Specker and the Pusey–Barrett–Rudolph reality theorems — are developed in this section, each as a chain of reasoning that runs entirely on results already established in the framework's technical papers: the Kochen–Specker case on the framework's equivalence with quantum mechanics (Main paper), and the PBR case on that equivalence together with the spatial-Markov and tensor-product structure of Main §3.3–3.4. A reader, and especially a referee, may treat all four subsections as developed; the two developed here are flagged case by case so the referee can check the chain.*

Bell's theorem. Bell's theorem shows that no local hidden-variable theory with a factorizable joint distribution can reproduce the quantum correlations. The framework is a hidden-variable theory — a completion of quantum mechanics in exactly the sense the EPR argument demanded [EPR], which is why Bell's theorem binds on it with full force — the substratum configuration is the hidden variable — and it reproduces the correlations, so it must fail an assumption of the theorem. The framework's Main paper establishes which: factorizability fails, while locality is retained. Two subsystems interacting at preparation carry a joint substratum transition matrix that does not factorize into a product of single-subsystem matrices; the marginalized stochastic process is, in the framework's terminology, P-indivisible, admitting no well-defined intermediate conditional probabilities of the kind Bell's factorization requires. Locality, meanwhile, is retained structurally: on the bounded-degree coupling graph with range-one coupling, space-like separated subsystems with disjoint coupling neighbourhoods are conditionally independent given boundary data, which delivers parameter independence with no fine-tuning. The non-factorizability is outcome-dependence in the Jarrett decomposition, not parameter-dependence. *(Established: framework Main paper.)*

Cast in the levels language of Section 13, with a precision the summary above needs. At fixed total microstate, a deterministic and causally local dynamics makes outcomes functions of settings and local past cones, so factorizability *at fixed* λ is a theorem of determinism plus locality — not an assumption a deterministic-local account is free to deny. What the framework declines to supply is the *ensemble* Bell's factorization integrates over: the fixed-λ, varied-settings counterfactual ensemble, which a closed deterministic universe does not contain, the settings being functions of the total state. This is the same move the Frauchiger–Renner case below makes explicit — the theorem's premise is uninstantiable rather than violated; the framework's structure does not contain the object the assumption is about. The operational form of measurement independence — the testable statement about actual ensembles of runs — holds, and holds dynamically rather than by stipulation: the substratum's measured chaotic mixing destroys residual correlations between pair-preparation and setting-generating degrees of freedom. That is also the precise sense in which the framework is not superdeterministic: the λ–setting correlations a superdeterministic explanation would need to do the work are here dynamically suppressed, and the correlations are produced by the indivisible joint dynamics of the emergent process, at exactly the Tsirelson bound. The theorem is met at the correct level rather than evaded: the emergent process is a non-factorizable stochastic process — which is what Bell's theorem says a successful account must provide — and the substratum level declines the theorem's premises rather than violating them. *(Established: framework Main paper §3.3, including the explicit substratum-level statement.)*

The Frauchiger–Renner theorem. Frauchiger–Renner shows that quantum theory cannot consistently describe its own use by an agent who is themselves a quantum system, if one assumes a single universal quantum description applying to all agents at once. The framework's Main paper addresses this through its treatment of the Wigner's-friend structure. In the framework, observer identity is partition-gauge: Friend and Wigner correspond to different choices of the visible/hidden partition of the same substratum, and each choice yields its own internally consistent reduced description. Friend's claim to a definite outcome and Wigner's description of Friend as a traced-out subsystem are statements in different reduced descriptions, not competing claims about the substratum. At the substratum level all parties have definite configurations; the apparent paradox is a covariance property — predictions are covariant under choice of observer partition.

In the levels language: Frauchiger–Renner assumes a single universal quantum description. The framework denies exactly this at the relevant level — there is no single quantum description, there are multiple partition-relative reduced descriptions, each obtained by a different trace-out of the one substratum. The universal description the theorem needs as its target does not exist in the framework; what exists universally is the substratum, which is not a quantum description at all. The theorem's assumption fails not by contradiction but because the framework's structure does not contain the object the assumption is about. *(Established: framework Main paper.)*

The Kochen–Specker theorem. Kochen–Specker shows that no noncontextual hidden-variable assignment — one assigning every observable a definite value independent of which compatible set of observables is measured alongside it — can reproduce quantum mechanics. The framework's relation to it is established by a short argument on a result the framework already has. The framework reproduces quantum mechanics: the Main paper's characterization theorem is a full equivalence between the embedded-observer reduced description and unitary quantum mechanics. Kochen–Specker is a theorem *about* quantum mechanics — it states that the observable algebra of quantum theory in dimension three or higher admits no noncontextual value assignment. Any account that reproduces quantum mechanics therefore reproduces a structure with no noncontextual valuation; the framework's valuation of the emergent observables is necessarily contextual, on pain of contradicting the framework's own equivalence theorem. Contextuality is thus not a property the framework must be argued into — it is forced by the equivalence.

What the framework adds is the *account* of that contextuality, and here the Section 13 architecture does the work. An emergent observable's value is produced by the trace-out, and the trace-out is taken with respect to a partition; measuring a compatible set of observables is a specific apparatus coupling, and the framework models a measurement context as exactly such a coupling-and-partition configuration (the double-slit treatment of Main §3.4 is the worked instance: a detector couples the system to additional visible-sector degrees of freedom and thereby changes the transition matrix). Which emergent observables are jointly defined, and the value each takes, are features of the reduced description fixed by that partition — not partition-independent functions of the substratum configuration. Contextuality, in the framework, *is* the partition-dependence of emergent valuation: the same partition-relativity Part II identified as gauge. The framework need not reconstruct the Kochen–Specker colouring obstruction itself; that obstruction is inherited intact with quantum mechanics. The framework is a deterministic hidden-variable theory whose hidden level is a definite substratum configuration but whose emergent valuation is contextual — precisely the kind of hidden-variable theory Kochen–Specker leaves open. *(Established: this paper, on the Main paper's equivalence theorem.)*

The reality theorems (Pusey–Barrett–Rudolph family). The PBR theorem shows that, given a natural assumption about how independently prepared systems combine — the Preparation Independence Postulate, that the joint ontic-state distribution of two separately prepared systems is the product of the single-system distributions — the quantum state cannot be a mere probability distribution over deeper states of reality, but must be ontic. A deterministic-substratum framework is naturally read as making the quantum state epistemic: the state is the observer's representation of incomplete knowledge of the substratum configuration. The framework's relation to PBR turns on the same level distinction as the other cases, and is established on results the Main paper already contains.

The preparation-independence the postulate requires is *unconditional* product independence. The framework provides product structure for separated systems — but conditionally, not unconditionally. By the spatial Markov property of the bounded-degree, range-one coupling graph (Main §3.3), space-like separated subsystems with disjoint coupling neighbourhoods are independent *given boundary data*; correspondingly the emergent description factorizes, $\mathcal{H}_{V_A} \otimes \mathcal{H}_{V_B}$ with $\rho_{AB} = \rho_A \otimes \rho_B$, but — as Main §3.4 states explicitly — only conditioned on boundary data, with the systems' correlation arising from shared boundary history. Conditional independence given a non-trivial shared variable is precisely *not* the unconditional product independence the Preparation Independence Postulate asserts; it is a common-cause structure, and the common cause — co-embedding in one substratum, with shared boundary data — is present at preparation and requires no interaction between the systems. PBR's no-go applies to models that satisfy the postulate; the framework is not one. The trace-out is again the mediator: marginalizing the shared boundary data is exactly what carries the emergent conditional factorization into substratum non-factorization. Note that this argument does not depend on classifying the framework as ψ-epistemic in PBR's single-level sense — PBR constrains models that are ψ-epistemic *and* postulate-satisfying, and the framework's failure of the postulate places it outside the theorem's scope regardless. *(Established: this paper, on the Main paper's equivalence theorem and the spatial-Markov and tensor-product structure of Main §3.3–3.4.)*

15. WHY THE UNIFICATION IS NOT AD HOC

The natural objection to any framework that "evades" several no-go theorems is that the evasions are opportunistic — that the framework has simply looked, for each theorem, at which assumption can be broken cheapest, and broken it. Part III's claim is that this objection does not apply, and the reason is structural rather than rhetorical.

The four cases are not four chosen assumptions. They are four manifestations of one feature: the framework has two levels of description, a hidden deterministic substratum and an emergent quantum description, related by the trace-out, and each no-go theorem's assumption runs aground on the separation that trace-out enforces. In three cases — Bell's factorizability, Frauchiger–Renner's universal description, PBR's preparation-independence — the property holds at the emergent level, where it is unobjectionable because that level simply is quantum mechanics, and fails at the substratum level, where the theorem needed it. In the fourth — Kochen–Specker's noncontextual valuation — the property holds at neither level, because the trace-out interposes a partition-dependent step that makes a noncontextual valuation of the emergent observables unavailable in principle (Section 13). The common factor is not a single "emergent-yes, substratum-no" slogan applied four times; it is the architecture — two levels and a trace-out — that, fixed before any no-go theorem is considered, determines the response to all of them at once. The framework did not select four assumptions to break; it has one structure, and the four responses are four things that structure does.

The test of non-ad-hocness is whether the response to the theorems was available before the theorems were examined, or constructed afterward to fit. Here it was available before: the two-level trace-out architecture is the framework's account of how quantum mechanics emerges at all — it is Part I's and Part II's subject, the construction itself — and it was not built to answer Bell or Frauchiger–Renner. That the same architecture, unmodified, answers the no-go family is a consequence drawn from it, not an adjustment made to it. This is the methodological content of the per-theorem evasion claim: not that the framework escapes four theorems, but that it does not have four escapes — it has one structure, and the four "escapes" are one fact about that structure, viewed four times.

The honest status, restated: the Bell and Frauchiger–Renner cases are developed in the framework's Main paper and summarized here faithfully; the Kochen–Specker and PBR cases are developed in Section 14 of this paper, each as a chain of reasoning that runs on results the framework's technical papers already establish — the equivalence with quantum mechanics for the Kochen–Specker case, and that equivalence together with the spatial-Markov and tensor-product structure of Main §3.3–3.4 for the PBR case. All four cases are thus developed results rather than conjectures, though two are developed here rather than in the technical papers, and a reader is invited to check those two chains directly. The methodological claim of Part III stands on all four: the framework's relation to the no-go family is governed by one architectural feature, and each of the four cases exhibits it.

PART IV — PHYSICS MODULO GAUGE

Part IV draws the three preceding commitments into a single methodological proposal: that the well-posed question for fundamental physics is not "what is fundamental?" but "what is the equivalence class of structures that produce the observations, and what are its representatives?" This part argues the proposal as a general methodology, not as a feature peculiar to the framework's particular construction.

16. THE QUESTION THAT DOES NOT CLOSE

The realism debate in the philosophy of physics is organized around a question: what is fundamental? Realists answer with an inventory — particles, fields, spacetime points, a wavefunction on configuration space — and divide over which inventory is correct. Anti-realists decline the inventory and divide over what to say instead. The debate is durable, and its durability is itself a datum. A question that has resisted resolution for as long as this one has — through every major change in physical theory, with the candidate inventories turning over each time while the debate's structure persists — invites the suspicion that the question is not hard but ill-formed.

The three preceding parts suggest where the ill-formedness lies. Part I found that "observation occurs," the framework's apparent single foundation, was not one claim but two, and that stating it at the right granularity meant resolving it into an indubitable axiom and a separable posit. Part II found that the structure consistent with the empirical inputs is unique, but only up to a gauge equivalence — and that the gauge freedom is not noise to be discarded but an exactly classified four-family group, the precise specification of which differences between descriptions are differences that make no difference. Part III found that the major no-go theorems turn on properties that are well-defined at one level of description and not at another, and that conflating the levels is what makes the theorems look like paradoxes.

The common thread is that each apparent puzzle was a failure to track a distinction the framework's structure makes precise: between what is committed and what is evidential, between what is structural and what is gauge, between what holds of the emergent description and what holds of the substratum. "What is fundamental?" fails to track the first of these in the same way. It presupposes that among the structures consistent with the evidence there is a fact of the matter as to which is *the* real one — that the gauge freedom Part II classifies is not gauge after all but conceals a genuine further fact. The reconstruction and gauge-classification theorems say there is no such further fact: the equivalence class is fixed by the evidence, and within it the choice of representative is, provably, without empirical content. "Which representative is the real one?" is not a hard question with an unknown answer. It is a question the framework's own structure certifies as empty.

17. THE WELL-POSED REPLACEMENT

If "what is fundamental?" is ill-posed, it should be replaced, not merely abandoned. The replacement the framework's methodology proposes is: *what is the equivalence class of structures that produce the observations, and what are its representatives?*

This question is well-posed in the precise sense that it has a determinate answer of a known form. The reconstruction theorem gives the equivalence class; the gauge-classification theorem gives the representatives, by enumerating the transformations relating them. The question decomposes into two sub-questions each of which is answerable by a theorem rather than by a stance: a uniqueness question (what class?) and a classification question (what gauge freedom within it?). Where "what is fundamental?" asks for a winner among the representatives and the framework certifies that there is none to be had, "what is the equivalence class and what are its representatives?" asks for exactly the two things the framework's theorems deliver.

Call this posture *physics modulo gauge*. The name is chosen to make the structure explicit. Physics, on this view, determines its content up to gauge — up to a classified equivalence — and the determination *up to gauge* is not a limitation to be apologized for but the complete and correct form of the answer. The realist's mistake is to want physics modulo nothing, a single representative crowned; the anti-realist's mistake is to conclude from the absence of a crowned representative that physics delivers no structural reality at all. Physics modulo gauge is the position that the equivalence class is real, fully and determinately, and that the gauge freedom is real too — real *as gauge*, that is, as the exactly specified set of distinctions that carry no empirical content. Neither the class nor the gauge freedom is a defect. Together they are the answer.

The posture has affinities with existing positions, and locating it among them sharpens it. Healey's analysis of gauge theories [Healey] argues that gauge freedom is not mere descriptive redundancy to be quotiented away without loss — that there is interpretive content in how a gauge theory's surplus structure is handled; physics modulo gauge agrees that the gauge freedom is not nothing, and adds that, when the gauge group is *exhaustively classified* as in Part II, the freedom becomes a determinate object of knowledge rather than an interpretive residue. Wallace's "mathematics-first" formulation of structural realism [Wallace2022] — on which one states the physics in its mathematical form and reads the ontology off that, rather than first fixing an object ontology — is the variant of structural realism closest to the present posture; physics modulo gauge can be seen as a mathematics-first structural realism in which the "mathematics" is specifically the equivalence class fixed by the reconstruction theorem and the gauge group classifying its representatives. The difference from the programmatic versions, as Part II stressed, is constructiveness: the class and the group are exhibited, not promised.

The posture reframes several standing puzzles, beyond the realism debate itself. The "unreasonable effectiveness of mathematics" becomes, in this setting, a theorem rather than a mystery: if the mathematical structure and the physics determine each other up to gauge given the structural assumptions, the effectiveness is what the reconstruction theorem asserts, and the residual sense of mystery attaches only to the structural assumptions themselves, which are stated and finite. The question whether mathematics *is* reality or merely *describes* it is, *for the embedded physical observer*, exhibited as gauge — the two phrasings are related by a transformation that changes no observable an embedded observer can perform, so the question of which is correct has, for physics, the same status as the question of which substratum representative is the real one. This is gauge *in the framework's specific physics-internal sense*: the framework's gauge classification is relative to what an embedded observer can measure, not relative to all possible vantage points. The framework does not claim the question is absolutely meaningless. Whether the question has substantive content in mathematical foundations, meta-mathematics, or other modes of inquiry not bound by embedded physical observation is a separate matter the framework does not address; what the framework establishes is that the question carries no empirical weight for physics. A specialist in mathematical foundations is free to pursue it; the framework's reconstruction theorem provides, modulo gauge, the specific mathematical structure that would be in question if it has a positive answer, without committing to whether it does. The measurement problem, on this posture, is not solved by adding a mechanism but dissolved by noting that the definiteness of outcomes is a feature of every partition-relative emergent description, and that the apparent conflict between descriptions is the partition-gauge covariance of Part III.

A further structural observation, kept brief here because it lies outside the framework's primary scope: if a mathematical-foundations specialist pursues the question and adopts a Tegmark-style position that all mathematical structures have ontological status, the framework's reconstruction theorem provides a natural *selection principle* within that class. The structures that admit embedded observers satisfying the framework's structural conditions (C1–C4 of the framework's Main paper, plus the gauge classification developed in Part II) form a precisely characterizable sub-class — the "observer-supporting" structures. This addresses what is sometimes called the measure problem for mathematical-universe ontologies: given that all mathematical structures are taken to be real, why does our universe instantiate the specific structure it does? The framework's answer, available to anyone pursuing the question in that direction without requiring commitment to the underlying ontology: the structures we can observe are precisely the observer-supporting ones, characterized as a selection effect. The framework does not commit to a mathematical-universe ontology; it provides the selection principle that any such ontology would benefit from having if it is pursued.

18. THE PROPOSAL AS A GENERAL METHODOLOGY

The claim of this part is not merely that the Incompleteness framework happens to admit a physics-modulo-gauge reading. It is that physics modulo gauge is a methodology — a way of posing and answering foundational questions — that other programs in fundamental physics could adopt, and against which they can be measured.

The proposal can be located against a long-standing open problem. Hilbert's sixth problem [Hilbert] called for the axiomatization of physics — the treatment of physics, in those parts ready for it, with the rigour of an axiomatic mathematical theory. More than a century on, the problem is generally regarded as only partially addressed, and a recognized obstacle is that the primitive notions of physical theories — "body," "force," and their successors — tend to be, in the phrase of the problem's modern commentators, pregnant with theory rather than genuinely primitive. The methodology proposed here bears on that obstacle directly. Its discipline is to identify the genuine primitives by stripping the theory-laden ones away — Part I's distillation of "observation occurs" to a two-axiom base is exactly this move, performed at the foundational layer — and then to ask, of the structure those primitives plus the empirical inputs determine, the two well-posed questions of physics modulo gauge. The proposal is, in this sense, a partial and specific response to Hilbert's sixth problem: not an axiomatization of physics entire, but a method for stating what a particular foundational program commits to, with the primitives exposed and the gauge freedom classified.

Stated generally, the methodology has three steps. First, for a candidate fundamental theory, identify the empirical inputs and ask whether they determine a structure uniquely — whether a reconstruction result is available. Second, where uniqueness holds only up to some equivalence, classify the equivalence explicitly: exhibit the gauge group, and prove the classification exhaustive, so that the surplus is bounded and known. Third, having the class and its gauge group, decline the question of which representative is real, and certify that declension with the gauge classification itself — the representatives are inter-transformable by observable-preserving maps, so the question has no empirical content.

A program that carries out these three steps has, on this view, said everything there is to say about what its theory commits to. A program that cannot carry out step one — that cannot exhibit its structure as determined by the evidence — has not yet earned a realist reading of that structure, whatever its other merits. A program that carries out step one but not step two — that has a structure but no classification of its gauge freedom — has an unbounded surplus and cannot yet say which of its commitments are genuine. The methodology is, in this sense, a standard: it specifies what a foundational program must exhibit before its ontological claims are well-posed, and it does so in a way that is checkable, because each step is the kind of thing that is done by a theorem or not done at all.

This is the sense in which physics modulo gauge is offered as a contribution beyond the framework. The framework is one worked example — a case where the three steps have been carried out, with the reconstruction theorem for step one, the gauge-classification theorem for step two, and the certified declension for step three. But the steps are statable without reference to the framework's particular substratum, and a different program with a different ontology could carry them out differently. What the methodology asserts is that *until* they are carried out, in some form, the question "what is fundamental?" remains ill-posed for that program — and that once they are carried out, the question has been replaced, not answered, by the two well-posed questions that physics modulo gauge substitutes for it.

The honest scope of the claim: that physics modulo gauge is the *correct* general methodology is not established by one worked example, and Part IV does not claim it is. What the framework establishes is weaker and still substantive — that the methodology is *instantiable*, that there exists at least one foundational program in which the three steps are carried out and the realism question is thereby reframed rather than left open. Whether the methodology generalizes — whether other foundational programs can or should be held to the same three steps — is itself left as the proposal's open horizon, in the same spirit in which Part I left the succession no-go open to a derivation that has not been found. The methodology is offered for use and for criticism, not asserted as proven.

19. GAUGE FREEDOM AND OBSERVATIONAL INCOMPLETENESS

This section is the most recent addition to the paper and the least settled; it is developmental even by the standards of the rest, and is offered as a proposal for scrutiny. It connects the methodology of Parts II–IV to the framework's core physical thesis — the incompleteness of observation — and argues the two are, on a precisely delimited class, one principle.

The framework takes its name from a thesis in physics: that an observer, being a proper part of the system it observes, cannot fully resolve that system — the trace-out over the unobserved remainder destroys information that no embedded observer can recover. Observational incompleteness, so stated, sounds like a limitation: a deficit, a wall past which the observer cannot see. Parts II through

IV have been about something that sounds quite different — gauge freedom, the *surplus* in a description, the respects in which distinct representations say the same thing. A deficit and a surplus appear to be opposites. This section argues they are not — that on a precisely identified class of cases, gauge freedom and observational incompleteness are the same fact seen from two sides, and that recognizing this is what lets the methodology of physics modulo gauge connect to the framework's foundational physics rather than sit beside it.

Begin with the apparent opposition, because dissolving it is the crux. Gauge freedom is descriptive redundancy: two representations, no observable difference, the surplus discardable. Observational incompleteness is missing information: a real system, an embedded observer, a fact the observer cannot reach. Surplus versus deficit. But ask *why* a gauge difference is surplus. The state-relabelings of the framework's substratum gauge group are surplus precisely because no observation distinguishes the relabeled substrata. The redundancy is not a free-standing fact about the representation; it is *constituted by* the observer's inability to tell the representatives apart. "There is surplus structure here" and "the embedded observer cannot access this difference" are not opposed claims — they are one situation, described from the representation side and from the observer side. The surplus *is* an inaccessibility; the inaccessibility *is* what makes it surplus.

This identification must be delimited, because not every observational incompleteness is gauge, and the cases that are not are as instructive as the cases that are. Consider two ways the framework's embedded observer fails to have a fact.

In the first, there is no fact to have. The exact extent of the hidden sector, the labels on substratum states, the question of whether the underlying contrast is tenseless or temporal (the subject of this paper's foundational Part I) — for each of these, the framework's own gauge classification establishes that *no observable depends on the answer*. There is not a hidden truth the observer is failing to reach; there is no truth there to reach, because a difference that no observation could ever register is not a difference in fact. Call this **gauge incompleteness**: the observer's picture is incomplete in the bookkeeping sense that a question can be posed, but the incompleteness is empty, because the question has no answer-bearing content. This species of incompleteness *is* gauge freedom — the name records the identification, which is a result and not a stipulation.

In the second, there is a definite fact, and the observer still cannot reach it. The observer's own future state, under the deterministic substratum dynamics, is fixed — there is a fact of the matter about it. But to compute that future the observer would need to be a system at least as large as the whole of which it is a proper part, and it is not; it is $V \subsetneq S$. Here the incompleteness is genuine and the missing item is a real fact, withheld not by the absence of content but by the observer's finite computational reach. Call this **computational incompleteness**. It is *not* gauge — a larger computer, still inside the same physics, could reach the fact; nothing about it is surplus.

The distinction between these two species is, as far as we are aware, not drawn in the existing literature on observational limitation, and drawing it is part of this section's contribution. With it in place, the identification can be stated exactly. *Gauge freedom is identical to gauge incompleteness — the species of observational incompleteness in which the inaccessible question has no answer-bearing content.* It is not identical to computational incompleteness. "Physics modulo gauge," the methodology, and "observational incompleteness," the physics, name one principle restricted to the gauge-incompleteness species: the methodology-side name and the physics-side name of the same fact.

One objection must be met head-on, because it is the natural one for a reader versed in the philosophy of gauge theories. It will be said that "surplus structure" is contested — that there is a body of work (Nguyen, Teh, and Wells [NguyenTehWells]) arguing surplus structure is not eliminable but essential, and that an identification of gauge with a *species of incompleteness* cannot be cleaner than the contested notion it leans on. The reply is that the contest is real but does not touch this identification, and the reason is a distinction drawn by Bradley and Weatherall [BradleyWeatherall]: "surplus structure" carries two nearly opposite senses. In one — *formulational* surplus — the structure is what a theory needs in order to be written as a local field theory; this is the sense in which surplus is essential, and the Nguyen–Teh–Wells result concerns it. In the other — *interpretational* surplus — the structure is descriptive redundancy with no fact of the matter as to which representative obtains. The framework's gauge — the substratum gauge group, classified by the gauge-classification theorem, with the equivalence class taken as what is real — is interpretational surplus, unambiguously: the theorem classifies transformations that preserve every observable. The "surplus is essential" result lives in the formulational sense and does not transfer. The identification of gauge with gauge incompleteness inherits no controversy it has not earned; it concerns interpretational surplus, and for that sense the "no fact of the matter" characterization is exactly standard.

There is a consequence of this identification that is worth stating in closing, because it is the point at which the paper's methodology turns back on the paper's own first part. If gauge freedom is the gauge-incompleteness species of observational incompleteness, then physics modulo gauge — the methodology — is a methodology *for working in the presence of observational incompleteness.* And observational incompleteness, in this framework, is not a special circumstance; it is the condition of any embedded observer, which is to say of any observer at all. The methodology is then not a special technique for a special class of theories. It is the general form that honest physics takes once it is admitted that the physicist is inside the system. Part I found that the framework's own foundation, examined closely, has a feature — whether the underlying contrast is tenseless or temporal — that is gauge: a question the framework can pose and that no observation it could ever make will answer. That the framework's *own foundational description* contains gauge incompleteness is not an embarrassment to be hidden. It is the methodology applying to itself, exactly as it should, if the methodology is correct: the physicist is inside the system even when the system in question is the physicist's own foundation.

The gauge-incompleteness / computational-incompleteness distinction admits a wider reading worth flagging, since it connects this paper's methodology to a strand in recent epistemology. Deutsch [Deutsch] argues in *The Beginning of Infinity* that knowledge itself is structurally incomplete — that from any current state of explanatory understanding there is always more knowledge to be created, and this incompleteness is a constitutive feature of what knowledge is rather than a contingent limitation awaiting some final theory. The distinction the present section draws between two species of observational incompleteness aligns naturally with a distinction Deutsch's account requires though does not state in these terms: gauge incompleteness is *structural* — the question has no answer-bearing content because no observation could distinguish — and computational incompleteness is *contingent* — the answer exists but requires more capacity than the observer has. The structural form is the one the framework's gauge-classification theorem identifies; the contingent form is the one Deutsch's account is built around. Both are real species of incompleteness; only the first is gauge.

The connection has a direction. Deutsch's argument is independent of any specific physics — it argues from explanatory creation being open-ended — but the framework provides one physics-foundational route to grounding it. Knowledge derived from observation cannot reach what observation cannot reach; the gauge-classification theorem establishes that observation, structurally, cannot reach the substrate's gauge equivalence class; therefore knowledge built from observation must inherit observational incompleteness as its lower bound. The reverse implication does not hold automatically — BoI does not presuppose this framework's specific machinery. What the framework adds is a stronger and more specific grounding than BoI provides on its own: knowledge is incomplete not only because explanatory creation is open-ended, but also because observation, the source of empirical input, is itself structurally incomplete in a way the gauge-classification theorem makes precise. The methodology of this paper is therefore one specification of what Deutsch's epistemological commitment requires of any foundational program that takes observation seriously: identify the gauge incompleteness, exhibit it constructively, and decline the questions that fall inside it as ill-posed rather than unanswered.

PART V — THE OPERATIONAL METHODOLOGY

Part V develops the methodology of the preceding parts at the operational level: given the substratum gauge group classified by Part II, what does substrate-only computation determine, and what does it provably not determine? The operational answer turns out to be non-trivial in both directions, with two no-

go theorems and the systematic audit results developed in the technical papers serving as the concrete demonstrations. Part V is more recent than Parts I–IV and is offered as developmental in the same sense the earlier parts are — provisional, but with each section's specific developmental status flagged.

21. THE OPERATIONAL QUESTION

Parts II through IV established that physics modulo gauge is the right *form* of the answer for a foundational program with a constructively classified gauge group. The well-posed questions, on that account, are what the equivalence class produces and what its representatives are; the ill-posed question is which representative is real. What Parts II–IV did not yet ask, and what Part V develops, is the operational form of the same question: given that the equivalence class is fixed by the reconstruction theorem and the gauge group classified by the gauge-classification theorem, which quantities are determined by substrate-only computation, and which depend on the choice of representative inside the equivalence class?

The question is non-trivial in both directions. Some quantities the framework would *like* to derive from substrate computation alone turn out to depend on choice of representative; the operational methodology forces these to be classified differently than the framework's first-pass framing would suggest. Other quantities turn out to be substrate-derivable in ways that the framework's first-pass framing did not make explicit; the operational methodology surfaces these as content that had been undercounted. Part V develops both directions through two no-go theorems (§22, §23), the architectural pattern that unifies them with Part III (§24), the systematic audits the technical papers carry out (§25), and the methodological refinements the audits canonize (§26).

The status of Part V as a whole is the same as that of Parts II–IV: the methodology is a proposal, the worked instances are developed results in the framework's technical papers, and the unification stated here as synthesis is the present paper's contribution. Each section flags its own status more specifically.

22. THE GAUGE-INVARIANCE NO-GO

Status: developed result; the theorem and its proof are in the framework's technical record. The statement and proof sketch below are faithful to that source; the methodological reading is the present paper's synthesis.

The first operational result is a no-go theorem at the level of substrate computation. State it sharply. Let $\mathcal{G}_{\mathrm{sub}}$ denote the substratum gauge group classified in Part II (the group of substratum transformations that preserve every observable accessible to an embedded observer). Let a *substrate-only computation* be any procedure that operates on a single representative of the substratum equivalence class and produces a numerical output. Let a *bijection-specific quantity* be any quantity whose value depends on which representative is chosen within the equivalence class — equivalently, any quantity that is not $\mathcal{G}_{\mathrm{sub}}$-invariant. The theorem states that no substrate-only computation determines a bijection-specific quantity.

The proof is short. By the gauge-classification theorem (Part II), the equivalence class $[(S, \varphi)]/\mathcal{G}_{\mathrm{sub}}$ is the framework's content; every observable depends only on equivalence-class invariants. A bijection-specific quantity, by definition, takes different values on representatives related by $\mathcal{G}_{\mathrm{sub}}$. A substrate-only computation, by definition, operates on one representative. If the computation's output is a bijection-specific quantity, then the same computation applied to a $\mathcal{G}_{\mathrm{sub}}$-related representative produces a different output. But no observable depends on the choice of representative. The computation therefore produces an output that is not an observable — it produces a value, but the value has no empirical content. Substrate-only computation, on this account, can determine an equivalence-class invariant or it can produce a non-observable; it cannot determine a bijection-specific quantity that is also empirically meaningful.

The reading of the theorem is the methodologically important part. A bijection-specific quantity is not a forbidden quantity; it is a quantity whose value is selected by the empirical inputs that pick the representative, not by the substrate's structure. Three quantities the framework's empirical record contains illustrate the point. The directional content of the framework's Condition 2 sum rule — which pair of mixing-angle ratios receives the 4/3 coefficient — depends on the assignment of mass eigenstates to generations, an assignment made by experimental measurement rather than by substrate computation. The CP-violating phases of the CKM and PMNS matrices are bijection-specific in the same sense: the absolute phases require fixing a flavor basis, and the basis is selected by which physical particle is identified with which substrate state. The mass-hierarchy assignment within a generation — which of the two states in a pair is the heavier — is bijection-specific in the same way. Each of these quantities the framework reproduces in its empirical record; each does so by taking the experimental assignment as input rather than by deriving it from substrate computation alone.

The methodological lesson is that the boundary between substrate-derived content and empirically-input content is sharper than a framework's first-pass framing typically makes it. A claim that "the framework derives the 4/3 sum rule" is correct at the magnitude level (the coefficient 4/3 is an equivalence-class invariant, fixed by the cubic-group representation theory) and incorrect at the directional level (which pair-ratio gets the 4/3 is bijection-specific, fixed by the empirical mass-hierarchy assignment). Both statements are about the same prediction; the methodological discipline is to keep them separated. This separation is what the operational methodology of Part V proposes as the standard for any foundational program of this shape: state explicitly, for each prediction, what is gauge-derivable and what depends on the empirical inputs that pick the representative.

The theorem has a converse that should be stated, since it is the part that prevents the result from being mistaken for a deflationary one. Quantities that *are* $\mathcal{G}_{\text{sub}}$-invariant — the equivalence-class invariants — are determined by substrate computation, and the framework's empirical record contains many of them: the Cabibbo magnitude $\lambda = 1/(\pi\sqrt{2})$, the Koide ratio 2/3, the Bekenstein–Hawking entropy coefficient 1/4, the MOND acceleration scale $a_0 = cH/6$, the structural forms of the mixing angles. These are not bijection-specific; the substrate determines them. The gauge-invariance no-go does not deflate the framework's substrate-derivable content; it draws the line between that content and the content that requires empirical input. Where the line falls is the operational question of §22; the answer is not vacuous in either direction.

23. THE CATEGORY-ERROR FINDING

Status: developed result; the type-mismatch is established by direct inspection of the framework's substratum-level structure (see the framework's substratum paper, §2.5, and the technical-paper record more broadly). The methodological reading is the present paper's synthesis.

The second operational result is more pointed than the first. It states that some questions the framework's first-pass framing makes available are not merely gauge-forbidden but ill-typed: the question asks for an object the framework's substratum-level structure does not contain.

The instance through which the finding was identified is the substrate-level Lagrangian. The framework's quantitative content (mass ratios, mixing angles, coupling thresholds) takes Lagrangian form at the *emergent* level: post-trace-out, after the substratum dynamics have been marginalized over the unobserved hidden sector, the visible-sector dynamics admit a description as a quantum field theory with an action functional. The substrate-level dynamics, by contrast, are *not* given by an action functional. The substrate is the deterministic linear bijection $\varphi : S \to S$ of the foundational definition; its dynamics is the iteration of φ, not the extremization of an action. A "substrate-level Lagrangian" is therefore not a thing the framework's substratum-level structure offers; the construction has no place to attach.

The mismatch is type-level rather than calculational. It is not that a substrate-level Lagrangian is hard to compute or that the relevant calculations are beyond current tools; it is that the request itself does not pick out a well-typed object in the framework's foundational ontology. Asking for "the substrate Lagrangian" is structurally similar to asking for "the temperature of a single classical particle" — temperature is a thermodynamic quantity defined on ensembles, and applying it to a single particle is not a difficult question but a category error. The framework's substrate is the bijection on a finite set; Lagrangian field theory is the post-trace-out description that emerges from marginalizing the bijection over the hidden sector. The two live at different levels of description, joined by the trace-out, and the joining is one-directional: emergent Lagrangians come *from* the substrate by trace-out, but there is no inverse map that produces a substrate-level Lagrangian *from* the emergent one.

The methodological point generalizes beyond the Lagrangian case. Any framework with a level distinction — substrate and emergent, fundamental and effective, microscopic and macroscopic — will have constructions natural at one level that do not transfer to the other, and not all such non-transfers are blocked by gauge invariance in the §22 sense. Some are blocked by type-mismatch: the construction is not defined at the level it is being asked of. The methodological discipline is to recognize the difference. A gauge-forbidden quantity exists at the relevant level but cannot be picked out by substrate-only computation; a type-mismatched construction does not exist at that level at all. Conflating these blocks two different kinds of inquiry under the same label and obscures what the framework's substrate-level commitments actually are.

The category-error finding has a corrective consequence for the framework's own research program. The framework's technical record contains a sequence of attempts at substrate-level closure of certain quantitative predictions — attempts that constructed candidate substrate-level Lagrangians and looked for the framework's coefficients to fall out of those Lagrangians. The category-error finding establishes that such attempts were not failing for calculational reasons; the targets they aimed at were not well-typed objects in the framework's foundational ontology. Re-classifying the relevant quantities as either equivalence-class invariants (substrate-derivable in the sense of §22) or bijection-specific (selected by empirical input) — rather than as targets for substrate-Lagrangian closure — is the corrected research direction. The reclassification is internal to the framework, but the methodological pattern it instances is general: when a foundational program finds itself repeatedly stuck on a construction, the first check should be whether the construction is well-typed at the level it is being asked of.

24. THE ARCHITECTURAL PATTERN: PART III EXTENDED

Status: synthesis; the architectural pattern is articulated in Part III as a unification of the four no-go cases of quantum foundations. The extension to the §22 and §23 results is the present part's synthesis claim, offered as a proposal in the same sense Part III's unifying formulation is offered.

Part III argued that the framework's relation to the four no-go theorems of quantum foundations is not four separate evasions but one architectural feature applied to four distinct theorems. The feature is the substrate/emergent split joined by the trace-out: two levels of description, the substrate level deterministic and global, the emergent level the marginalization of the substrate over the unobserved hidden sector, and the trace-out the operation that produces the second from the first. Bell, Kochen–Specker, the Pusey–Barrett–Rudolph reality theorems, and Frauchiger–Renner each have an assumption that the framework declines, and in each case the declension is located by the substrate/emergent architecture rather than by a custom-built evasion for the theorem in question.

The §22 gauge-invariance no-go and the §23 category-error finding extend the same pattern. In each case, a question the framework's first-pass framing makes available — what substrate computation determines a bijection-specific quantity? what is the substrate-level Lagrangian? — is met not by a calculational answer but by a structural diagnosis of where the question lands relative to the substrate/emergent architecture. The gauge-invariance no-go locates the question at the substrate level and finds that the requested object is not an equivalence-class invariant, so substrate-only computation cannot supply it. The category-error finding locates the question at the substrate level and finds that the requested object is not defined at that level at all. The four Part III results and the two Part V results are six manifestations of the same architectural feature working in slightly different modes.

Stated as a thesis, the pattern is this: every claim the framework supplies and every claim the framework declines traces, ultimately, to the substrate/emergent architecture and the trace-out that joins the levels. The architecture is fixed before any specific question is considered, by the foundational structure of Part I and the gauge classification of Part II. The substantive content of the framework's no-go encounters — including both Part III's quantum-foundations cases and Part V's operational cases — is not a series of independent moves but a sequence of locations of where each question lands relative to that pre-fixed architecture. This is what "physics modulo gauge" looks like operationally: a methodology whose answers and declinations are determined by the architecture rather than chosen case by case.

The unification has a verifiable consequence. If the architectural pattern is right, then any further no-go encounter the framework runs into should also be locatable by the substrate/emergent split. The pattern is therefore falsifiable in a specific sense: a no-go result the framework cannot meet by locating where the assumption lands relative to the architecture would be evidence that the unification is incomplete and that a custom move is required. Part III and Part V have, to date, located every no-go encounter the framework has taken up by this single move; whether that record continues to hold under further scrutiny is part of what is offered for testing. The unification is a working claim, not a settled one.

25. THE OPERATIONAL AUDITS

Status: the audits themselves are developed in the framework's technical record; their methodological reading as systematic application of §22 to the framework's own content is the present part's synthesis.

The §22 gauge-invariance no-go is methodologically empty if not applied. A framework that proves a no-go theorem about substrate-only computation and then continues to claim quantities are substrate-derivable when they are in fact bijection-specific would have stated a result without using it. The framework's technical record applies the §22 result to its own content through systematic audits, which is part of what the operational methodology of this part means by "operational."

The audits proceed by going through each prediction the framework makes and asking, for each: is this quantity an equivalence-class invariant or bijection-specific? Where it is equivalence-class invariant, the framework's claim that it is substrate-derived stands. Where it is bijection-specific, the claim is corrected to reflect that the quantity is reproduced given empirical input rather than derived from substrate computation alone. The audits are not deflationary in net effect — the framework's substrate-derivable content is substantial after the audit is complete — but they are corrective on specific items, and the corrections matter for accurate characterization of what the framework claims.

Two audits in particular illustrate the discipline. A physics-sector audit covering the technical papers on the Standard Model and gravitational sectors classified twenty-nine predictions, finding approximately fifteen substrate-derivable (i.e., equivalence-class invariant), four substrate-derivable in magnitude with bijection-specific directional content, and ten emergent-level mechanism-distinctive items that the framework's first-pass framing had not separated from the substrate-derivable cluster. A cross-domain audit covering chemistry, biology, and computational complexity classified thirty-four cross-domain claims under the same criteria; a mid-audit correction caught a generous-grading pattern and revised the substrate-derivable count downward, separating mechanism-distinctive items from substrate-derivable ones in line with §22's distinction. The numerical findings of both audits are in the framework's technical record; what is methodologically important here is that the audits ran on the same standard the framework's operational methodology proposes for foundational programs in general.

A third audit applied the same standard recursively to a previous expansion of the substrate-derivable count. The expansion had identified eight further candidates not covered by the prior audits and reclassified them as substrate-derivable; the recursive audit found that three of the eight survive a strict-criteria filter as approximately partially independent substrate-derivable items (with correlation discounts reflecting that they share machinery with already-counted items), four are mechanism-distinctive in §22's sense (distinctive in the framework's *explanation* for outcomes that the framework shares with the standard account), and one is framing rather than predictive. The recursive correction is methodologically the cleanest of the three findings: a framework that catches its own corrective moves inflating, and walks the inflation back to what survives strict criteria, is exhibiting the discipline the operational methodology recommends to others.

What the audits demonstrate as methodology is the second-order claim of this part. The §22 gauge-invariance no-go states that substrate-only computation cannot determine bijection-specific quantities; the audits use this no-go to classify the framework's own predictions; the resulting taxonomy distinguishes substrate-derivable content (equivalence-class invariants) from bijection-specific content (selected by empirical input) and from mechanism-distinctive content (distinctive in explanation though not in prediction-as-such). The taxonomy is what the framework offers in place of an undifferentiated count of "predictions." It is also what the operational methodology proposes for any foundational program of the same shape: not a count, but a categorized inventory in which each item is labeled by its relationship to the framework's gauge group.

The audits' general lesson concerns what the methodology asks of its users. A framework that proposes physics modulo gauge as its methodology takes on an obligation to apply the methodology to its own claims, and to revise those claims where the operational classification requires it. The framework's technical record shows that obligation being met more often than not — including in the recursive case above, where the obligation extended to revising the framework's own audits — though not always in the first pass. The methodological refinements that this discipline canonizes are the subject of §26.

26. THE OPERATIONAL REFINEMENTS

Status: the refinements stated here are derived from the framework's audit experience and are offered at the generality the operational methodology requires. The discipline they specify is canonical within the framework's technical record at multiple-instance strength; whether it generalizes to other foundational programs is the proposal-status claim.

The audit experience canonizes three methodological refinements that the operational methodology recommends as standards for any foundational program of the kind Parts II–IV describe. Each is stated below at the generality the methodology requires, with the framework-specific instance noted for reference but not relied on for the generality of the claim.

First refinement: verify before recording. When a finding adjusts a framework's count of substrate-derivable content — whether upward (more items than the prior count suggested) or downward (fewer items than the prior count suggested) — the finding should be subjected to strict-criteria filtering before being recorded as the new count. Positive findings inflate at first pass: the natural pattern is to count items that look as if they should qualify, and to discover on closer inspection that several of them do not meet the strict criteria. The discipline of running the strict-criteria filter on every count-change finding cuts the inflation, often by a non-trivial fraction. The framework's audit record contains eight instances of this pattern across one session of work, at strengths sufficient to canonize the refinement at the framework's internal level. The general claim is that any foundational program whose methodology proposes a categorized inventory will face the same inflation pressure, and the same strict-criteria filtering is the corrective.

Second refinement: recursive scrutiny. A corrective finding — one that catches an inflation in a prior finding — is itself subject to the same inflation pressure.

The framework's audit record contains one instance of this case (a corrective finding that itself inflated the framework's count of substrate-derivable content by a factor of three at first pass, and that walked back to a strict-criteria count after a second-order audit). The refinement is provisional in the framework's own canonization scheme — second-order corrections are rare enough that one instance does not establish the pattern at the same strength as the first refinement — but the methodological point is general: self-correction does not stop at one pass, and the strict-criteria discipline applies recursively. A framework that catches the inflation in its own corrections is exhibiting the methodology more sharply than a framework that catches only the inflation in its original findings.

Third refinement: the observable/derivable distinction. A foundational program's open-question list is not a uniform category; it contains questions that fall into at least four distinct buckets. Some questions concern *observable* content the framework does not yet predict — open empirical questions in the standard sense. Some concern content that is *derivable* from the framework's existing structure but has not yet been derived, requiring further calculation rather than further structure. Some concern content that is *structurally permitted* (consistent with the framework's commitments) but whose specification is gauge — equivalent representatives produce different specifications, none of which has empirical content. Some concern content that is *structurally forbidden* — gauge-incompatible specifications, ruled out at the level of the framework's structure. The four buckets ask different things of further work, and mixing them in a single "open questions" list obscures what each kind of question requires for closure. The methodological refinement is to distinguish the four buckets explicitly. The framework's open-question list, when classified by this distinction, is shorter than the unclassified list suggests, because some apparent open questions resolve to "structurally permitted but gauge" — not closures, but acknowledgments that the question has no answer-bearing content.

The three refinements share a structure. Each takes a category that a foundational program's working vocabulary collapses — the category of "predictions," the category of "self-correction," the category of "open questions" — and disaggregates it into sub-categories that ask different things and admit different kinds of further work. The methodology's discipline, throughout Parts II through V, has been the disaggregation move: track what is committed, distinguish it from what is not, and at each level find that the distinction the puzzle was failing to track is what makes the puzzle tractable. The refinements of §26 are that move applied to the methodology itself: track what kind of operational claim is being made, distinguish derivation from inflation from declension, and the methodology gains the same precision it asks the framework's content to display.

27. CONCLUSION

This paper has articulated the methodology of the Incompleteness framework through five commitments, and it is worth stating, in closing, what holds them together.

Part I distilled the framework's foundation and found it was two axioms rather than one, related by a no-go: the foundation stated at the exact granularity of what it commits to, with the single non-evidential posit named and isolated. Part II found that the framework's structural realism is constructive rather than programmatic: the structure consistent with the evidence is exhibited and shown unique, and the gauge freedom is exhibited and shown exhaustively classified. Part III found that the framework's relation to the no-go theorems of quantum foundations is one architectural feature — two levels and a trace-out — and not four separate escapes. Part IV drew these together into physics modulo gauge: the proposal that the well-posed question for fundamental physics is the equivalence class and its representatives, not a single crowned fundamental. Section 19, within Part IV, identified that gauge freedom is one species of the framework's own foundational thesis — observational incompleteness — so that the methodology and the physics are, on that species, one principle, and the methodology is revealed as the general form physics takes for an observer inside the system it describes. Part V developed the operational form of the methodology: two further no-go results (gauge-invariance of substrate-only computation; the category mismatch of a substrate-level Lagrangian), the unification of these with Part III's four cases under a single architectural pattern, the systematic audits that apply the operational standard to the framework's own content, and three methodological refinements canonized by the audit experience.

The unifying thread is a single discipline: track precisely the distinction between what is committed and what is not. At the foundational layer this is the distinction between the indubitable axiom and the separable posit. At the level of structure it is the distinction between the equivalence class and the classified gauge freedom. At the level of the no-go theorems of quantum foundations it is the distinction between the substratum and the emergent description. At the operational level it is the distinction between equivalence-class invariants (substrate-derivable) and bijection-specific quantities (selected by empirical input), and between either of these and constructions that are not well-typed at the level they are being asked of. And as a general methodology it is the distinction between the well-posed questions physics can answer by theorem and the ill-posed question — which representative is real — that it cannot. In each case the methodological move is the same: find the distinction the puzzle was failing to track, make it precise, and the puzzle either resolves or is shown to have been ill-formed.

That discipline is replicable, which is the paper's reason for stating it separately from the framework's physics. The framework's quantitative claims — the Standard Model retrodictions, the entropy coefficient, the neutrino prediction — stand or fall on their own evidence, in the technical papers. The methodology stands or falls on whether the discipline just described is sound and general. The two assessments are independent, and a reader may find the methodology worth adopting whatever they conclude about the physics, or worth rejecting whatever they conclude about the physics. We have marked, throughout, where the paper's claims are developmental: the foundational distillation of Part I is recent and offered for scrutiny; the structural theorems of Part II are summarized from the technical papers at the fidelity of faithful synthesis; the four no-go cases in Part III are all developed results, two of them developed in this paper rather than in the technical papers and flagged for the referee accordingly; the generality of the Part IV methodology is offered as a proposal, not a result; the identification of gauge freedom with the gauge-incompleteness species of observational incompleteness, in Section 19, is offered explicitly as a proposal for scrutiny; and Part V is more recent still — its two no-go results are developed in the framework's technical record but their methodological reading as part of one architectural unification with Part III is offered as a synthesis, the systematic audits are themselves developed but the discipline they instantiate is proposed for use beyond the framework, and the three operational refinements are canonical within the framework's record at multiple-instance strength while their generality to other foundational programs is the proposal-status claim. What is not developmental is the discipline itself — track what is committed, distinguish it from what is not — and the claim that, applied to the foundations of physics, that discipline is what the framework's methodology consists in.

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This paper articulates, for a philosophy-of-physics readership, the methodological posture developed across the Incompleteness framework's technical papers. Part I develops the framework's foundational commitment; Parts II and III draw on structural results from the framework's substratum and main papers.

REFERENCES

[Main] Maybaum, A. (2026). *The Incompleteness of Observation: The Equivalence of Quantum Mechanics and Embedded Observation.* Zenodo. DOI: 10.5281/zenodo.19060318.

[SM] Maybaum, A. (2026). *The Standard Model from Cubic-Lattice Substratum Dynamics.* In the Incompleteness framework collection, Zenodo. DOI: 10.5281/zenodo.19060318.

[Substratum] Maybaum, A. (2026). *The Substratum: Reconstruction and Interpretation.* In the Incompleteness framework collection, Zenodo. DOI: 10.5281/zenodo.19060318.

[GR] Maybaum, A. (2026). *Emergent Gravitation in the Incompleteness Framework.* In the Incompleteness framework collection, Zenodo. DOI: 10.5281/zenodo.19060318.

[Barbour] Barbour, J. B. (1999). *The End of Time: The Next Revolution in Physics.* Oxford University Press.

[EPR] Einstein, A., Podolsky, B., & Rosen, N. (1935). "Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?" *Physical Review* 47, 777–780.

[Bell] Bell, J. S. (1964). "On the Einstein Podolsky Rosen Paradox." *Physics Physique Fizika* 1, 195–200.

[BradleyWeatherall] Bradley, C., and Weatherall, J. O. (2020). "On Representational Redundancy, Surplus Structure, and the Hole Argument." *Foundations of Physics* 50(4), 270–293.

[Deleuze] Deleuze, G. (1968). *Différence et répétition.* Presses Universitaires de France. English: *Difference and Repetition*, trans. P. Patton, Columbia University Press, 1994.

[Deutsch] Deutsch, D. (2011). *The Beginning of Infinity: Explanations That Transform the World.* Viking Press.

[FrauchigerRenner] Frauchiger, D., and Renner, R. (2018). "Quantum theory cannot consistently describe the use of itself." *Nature Communications* 9, 3711.

[French2014] French, S. (2014). *The Structure of the World: Metaphysics and Representation.* Oxford University Press.

[FrenchLadyman2003] French, S., and Ladyman, J. (2003). "Remodelling structural realism: quantum physics and the metaphysics of structure." *Synthese* 136, 31–56.

[Halvorson] Halvorson, H., and Müger, M. (2007). "Algebraic quantum field theory." In *Handbook of the Philosophy of Physics — Part A*, J. Butterfield and J. Earman (eds.), Elsevier, 731–922.

[Healey] Healey, R. (2007). *Gauging What's Real: The Conceptual Foundations of Contemporary Gauge Theories.* Oxford University Press.

[Hilbert] Hilbert, D. (1902). "Mathematical Problems." *Bulletin of the American Mathematical Society* 8, 437–479. (The sixth problem: the axiomatization of physics.)

[KochenSpecker] Kochen, S., and Specker, E. P. (1967). "The Problem of Hidden Variables in Quantum Mechanics." *Journal of Mathematics and Mechanics* 17, 59–87.

[Ladyman1998] Ladyman, J. (1998). "What is structural realism?" *Studies in History and Philosophy of Science* 29, 409–424.

[LadymanRoss] Ladyman, J., and Ross, D. (with Spurrett, D., and Collier, J.) (2007). *Every Thing Must Go: Metaphysics Naturalized.* Oxford University Press.

[LadymanLorenzetti] Ladyman, J., and Lorenzetti, L. (forthcoming). "Effective Ontic Structural Realism." *British Journal for the Philosophy of Science.*

[McTaggart] McTaggart, J. M. E. (1908). "The Unreality of Time." *Mind* 17, 457–474.

[NguyenTehWells] Nguyen, J., Teh, N. J., and Wells, L. (2020). "Why Surplus Structure Is Not Superfluous." *British Journal for the Philosophy of Science* 71(2), 665–695.

[PBR] Pusey, M. F., Barrett, J., and Rudolph, T. (2012). "On the reality of the quantum state." *Nature Physics* 8, 475–478.

[Ruetsche] Ruetsche, L. (2011). *Interpreting Quantum Theories.* Oxford University Press.

[SpencerBrown] Spencer-Brown, G. (1969). *Laws of Form.* Allen & Unwin, London.

[Wallace2022] Wallace, D. (2022). "Stating structural realism: mathematics-first approaches to physics and metaphysics." *Philosophical Perspectives* 36, 345–378.

[Wassermann] Wassermann, G. "A reductive account of 'before': deriving temporal succession." PhilSci-Archive preprint 20893.

[Whitrow] Whitrow, G. J. (1980). *The Natural Philosophy of Time*, 2nd edn. Clarendon Press, Oxford.